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A Risk-Aware Multi-Agent Decision Support System for Stock Trading on the Warsaw Stock Exchange: A Forecast-Combination Approach with Walk-Forward Evaluation

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The source code and data-preparation scripts used in this study are available upon reasonable request.

ABSTRACT We study a multi-agent decision support system for stock trading on the Warsaw Stock Exchange (WSE), ~~a~~ an underrepresented Central and Eastern European market~~underrepresented in the algorithmic trading literature~~. Five heterogeneous technical-analysis agents feed a transparent, confidence-weighted decision agent ~~grounded in~~ motivated by forecast-combination theory. The primary contribution is a rigorous and reusable evaluation framework ~~, demonstrated through an in-depth empirical study of this ensemble. The framework~~ that addresses four recurring weaknesses of backtested trading studies ~~:~~ - in-sample parameter selection, absent transaction costs, neglect of risk-adjusted performance, and lack of significance tests ~~. Evaluation uses~~ - demonstrated through an in-depth empirical study of this ensemble. Evaluation combines anchored walk-forward tuning~~of decision-agent parameters~~, calibrated commissions and slippage, Sharpe, Sortino, Calmar, and Martin ratios and the Ulcer Index, a volatility-matched comparison, bootstrap confidence intervals, ~~and decompositioninto bull, correction, and sideways regimes~~regime decomposition, leave-one-agent-out ablation, and sensitivity analyses over decision-agent weights, selection objective, and costs, plus an empirical audit of the confidence heuristic. On 435 held-out trading days for WIG20, the ensemble trails passive buy-and-hold on gross total return in a bull-dominated window but improves every risk-adjusted metric ~~considered~~ and, once scaled ex ante to benchmark volatility, total return as well; these differences are economically meaningful yet not statistically significant on a single sub-two-year window. Regime analysis attributes the profile to smaller losses in corrections and gains in sideways phases, offset by lower bull exposure. The same qualitative pattern appears in external validation on the FTSE 100 under UK-specific costs, and the ensemble remains competitive with ~~an LSTM directional baseline~~ LSTM and gradient-boosted-tree baselines under the identical protocol. ~~Ablation and cost-sensitivity checks confirm robustness.~~ The results support ~~theory-grounded~~ theory-motivated ensembles as risk-aware decision support rather than return enhancers.

INDEX TERMS Algorithmic trading, cross-market validation, decision support systems, forecast combination, multi-agent systems, risk-adjusted performance, walk-forward validation, Warsaw Stock Exchange.

I. INTRODUCTION

DECISION support systems and algorithmic trading occupy a central place in contemporary financial markets. They help both institutional and retail investors process large volumes of market information and translate that information into trading actions [1]. Over the last two decades, research on algorithmic trading has expanded from simple rule-based strategies [2] to machine-learning classifiers [3], [4], and more recently to deep reinforcement learning systems that learn trading policies directly from data [5]–[7]. In parallel, multi-agent systems (MAS) have been proposed as a natural way to organize this diversity of trading rules and models, so that independent agents can produce signals that a coordinator combines into the final decision [8]–[11].

Despite the breadth of this literature, four methodological weaknesses recur in empirical studies of trading strategies.

The first is *in-sample parameter selection*: when the best configuration of a strategy is chosen on the same data used to evaluate it, the out-of-sample performance is overstated [12], [13]. The second is the *omission of realistic frictions*, in particular commissions and slippage, which can erode a noticeable fraction of the gross return of high-turnover configurations. The third is the *neglect of risk-adjusted performance*: many studies compare a trading strategy to passive buy-and-hold purely on cumulative profit, without taking into consideration that an active strategy can produce a smoother equity curve at a lower unleveraged return. This weakness is particularly acute during sustained bull markets, where passive exposure mechanically dominates any strategy that is occasionally out of the market [14]. The fourth is the *absence of formal significance tests*: headline metrics are often reported as point estimates without confidence intervals or paired tests against the benchmark, even though trading-strategy metrics are known to be non-Gaussian and path-dependent [12], [15]. Beyond these methodological issues, the empirical evidence on multi-agent decision support systems is strongly concentrated on major markets in the United States, Western Europe, and Asia; the Warsaw Stock Exchange (WSE) - the largest exchange in Central and Eastern Europe (CEE) by capitalization - has received much less attention [16]–[18].

The present paper addresses these four methodological weaknesses in combination on markets that have received limited attention in the multi-agent trading literature, with WIG20 as the primary panel and the FTSE 100 as external validation. First, it proposes a multi-agent decision support system for large-cap index constituents that aggregates five heterogeneous signal agents through a transparent decision agent. The aggregator is interpreted through the lens of forecast-combination theory [19]–[21]: each signal agent is treated as an imperfect forecaster of the next-period directional move, and the per-signal confidence value is ~~interpreted as a~~ used as a heuristic, monotone proxy for the ~~inverse variance~~ reliability of that forecaster at that point in time; whether this proxy carries measurable information is itself tested empirically in the evaluation. Second, the evaluation follows an anchored walk-forward protocol in which only the decision-agent parameters are tuned on training data, whereas indicator hyperparameters are fixed at their literature-conventional values and all headline metrics are reported on a held-out test window that is never seen during parameter selection. Third, all backtests include market-specific frictions: proportional commission and one-way slippage calibrated to typical Polish retail brokerage conditions on WIG20, with a full sensitivity sweep over both quantities (Section V-B), and UK commission, slippage, and Stamp Duty Reserve Tax on the FTSE replication (Section VI-O). Fourth, the evaluation is centered on *risk-adjusted* performance rather than on gross total return: we report Sharpe, Sortino, Calmar, and Martin ratios, and Ulcer Index; a volatility-matched comparison that scales the ensemble ex ante to the buy-and-hold volatility; bootstrap confidence intervals and paired significance tests; and a regime decomposition into bull, correction, and sideways subperiods of the held-out window. An LSTM directional classifier and a gradient-boosted-trees (GBM) classifier, evaluated under the identical walk-forward protocol, costs, and portfolio simulator on both markets (Section V-D1), ~~provides a modern~~ provide two structurally different learning-based reference ~~point~~ points against which the transparent ensemble is compared.

We position the contribution as primarily methodological-evaluative and empirical rather than algorithmic. The aggregation rule is a transparent confidence-weighted score whose simplicity is a deliberate design choice: it keeps the decision layer interpretable and makes the mapping to forecast-combination ~~interpretation exact~~theory transparent, in contrast to opaque end-to-end learners. On this foundation the paper offers, first, a rigorous and reusable evaluation framework for multi-agent trading systems together with an in-depth empirical study of one instance of it on markets that have received little attention, and second, ~~an explicit~~ a conceptual grounding of the aggregator in forecast-combination theory that turns the ensemble from an ad-hoc heuristic into a testable variance-reduction mechanism, together with experiments that probe directly how far this theoretical analogy holds in the data. These contributions can be summarized as follows.

- A transparent multi-agent decision support artifact for blue-chip stock trading whose confidence-weighted aggregation rule, though intentionally simple, is ~~explicitly~~ conceptually grounded in forecast-combination theory, together with a taxonomy that maps each signal agent to a distinct methodological family.
- A reusable evaluation protocol, which is the central contribution of the paper, combining anchored walk-forward validation, a calibrated transaction-cost model, a cost-sensitivity sweep over commissions and slippage, a leave-one-agent-out ablation, a ~~comparison against an LSTM deep-learning baseline~~ sensitivity analysis of the decision-agent family weights that includes in-sample-optimized and inverse-variance-estimated alternatives, a sensitivity check of the walk-forward selection objective, an empirical validation of the confidence heuristic, comparisons against LSTM and gradient-boosted-tree baselines evaluated under the identical protocol, and paired bootstrap tests with confidence intervals.
- A risk-adjusted evaluation comprising Sortino, Calmar, Ulcer, and Martin ratios, a volatility-matched comparison against index buy-and-hold, and a regime decomposition of the test window into bull, correction, and sideways subperiods. This evaluation makes it possible to describe when the ensemble adds value and when it does not, instead of relying on a single scalar comparison.
- An empirical correlation analysis of the agent signals that directly illustrates the variance-reduction mechanism predicted by forecast-combination theory.

- Empirical evidence from two markets with different structures and cost regimes: the WSE, underrepresented in the algorithmic trading literature, and an external validation on the FTSE 100. On both, the system trails passive buy-and-hold on gross total return during bull-dominated windows but improves risk-adjusted performance and reduces drawdowns, with the volatility-matched comparison favoring the ensemble on total return. These differences are economically meaningful but not statistically significant on the available windows, and we frame them accordingly.

The remainder of the paper is organized as follows. Section II reviews the related literature in four streams. Section III introduces the theoretical background and states the three hypotheses tested in the experiments. Section IV describes the system architecture, the individual agents, and the decision agent, and presents the agent taxonomy. Section V presents the experimental setup, including the dataset, transaction-cost model, walk-forward protocol, baselines, risk-adjusted metrics, regime labeling, volatility-matched comparison, cost-sensitivity procedure, and bootstrap tests. Section VI reports the results of conducted experiments. Section VII interprets the results against the three hypotheses and links the empirical correlations to the variance-reduction prediction of forecast-combination theory. Section VIII discusses limitations and future work. Section IX concludes the paper.

II. RELATED WORK

This section reviews four main streams of literature that directly inform the contribution of the paper: algorithmic trading and technical analysis, multi-agent systems in finance, algorithmic trading on Central and Eastern European markets, and risk-adjusted performance evaluation. The organization into four streams is intended to clarify where the present paper sits in a crowded research landscape and why the combination of the four angles is novel.

A. ALGORITHMIC TRADING AND TECHNICAL ANALYSIS

Technical analysis uses indicators computed from price and volume to generate trading signals. Early work established that simple trading rules based on moving averages and trading ranges exhibit predictive power that is hard to reconcile with the strict form of the efficient markets hypothesis [2]. Subsequent studies formalized this debate with algorithmic backtests and statistical tests [22], and showed that several popular indicators (MACD, RSI, Bollinger Bands, among others) continue to produce information when combined with machine-learning classifiers [3]. Closely related to the ensemble design of the present paper, empirical work on cross-sectional momentum [23], [24] and time-series momentum [25] has shown that trend-following and mean-reversion signals carry orthogonal information about future returns. More recent work has shifted to deep learning, in particular to recurrent architectures for price prediction [4], and to deep reinforcement learning systems that learn trading policies end-to-end [5]–[7], [26]. Systematic reviews conclude that the evidence on technical analysis is mixed and depends strongly on the choice of market, period, and evaluation protocol [27], [28]. To situate the transparent forecast-combination design of the present paper against this learning-based stream, we include ~~an LSTM directional classifier in the style of [4] as a modern baseline,~~ two learning-based baselines evaluated under the same walk-forward protocol, transaction costs, and portfolio simulator as the ensemble: an LSTM directional classifier in the style of [4], and a gradient-boosted-trees classifier [29], [30], a model family that large-scale comparisons identify as among the strongest learners on tabular financial features [31], [32].

Two methodological criticisms recur across this literature. The first is backtest overfitting: when a large number of configurations are evaluated on the same data, the best configuration is biased upward and the out-of-sample performance is overstated [12], [13]. The second is the frequent absence of realistic frictions, such as commissions and slippage, which can erode a large fraction of the gross return of a high-turnover strategy. The present paper takes both criticisms seriously and incorporates them into the evaluation design.

B. MULTI-AGENT SYSTEMS IN FINANCE

Multi-agent systems provide a natural abstraction for trading, because different strategies can be encapsulated as independent agents that communicate through well-defined interfaces. Agent-based artificial markets have been used to study the emergence of liquidity, volatility, and price-discovery patterns from the interaction of heterogeneous traders [33], [34]. Multi-agent decision support systems for stock and currency markets have been proposed as early as two decades ago, typically with specialized agents that generate buy/sell recommendations and a coordinating component that aggregates them [8]–[11]. The A-Trader system [9], [10] is a particularly close ancestor of the present design, since it also uses multiple specialized agents and a consensus-based coordinator, but it has been evaluated primarily on FOREX data. Recent work has extended the MAS paradigm with multi-agent reinforcement learning for portfolio management [5] and algorithmic trading [6]. Outside finance, a substantial body of control-theory research studies how interacting agents reach agreement through consensus protocols, including consensus tracking over cooperation-competition networks [35], convergence-rate analysis under communication delays [36], and logic-based switching schemes for uncertain nonlinear multi-agent dynamics [37]. That literature analyses the dynamical convergence of agent states to a common value over time, whereas the coordinator studied here is a static forecast-combination rule that aggregates heterogeneous directional forecasts at each date; the two notions of agreement are related in spirit but formally distinct.

Relative to this literature, we do not claim the aggregation mechanism itself as novel: combining specialized agents through a weighted score is well established in the systems cited above. The distinction of the present work is twofold. First, it makes the conceptual connection between the aggregator and forecast-combination theory [19]–[21] explicit, interpreting the coordinator score ~~not as an ad-hoc heuristic but~~ as a weighted sum of heterogeneous ~~forecasts, and~~ directional forecasts, using this interpretation both in the design of the decision agent and in the motivation of the ablation and correlation experiments, and testing its limits empirically through the confidence-validation and weight-sensitivity analyses of Sections VI-H and VI-J. Second, and more importantly, it subjects the resulting ensemble to a rigorous evaluation protocol, described in Section V, that most prior multi-agent trading studies do not apply in combination. The intended contribution is thus the theoretical grounding and the evaluation, not the aggregation rule.

C. ALGORITHMIC TRADING ON CENTRAL AND EASTERN EUROPEAN MARKETS

Although the Warsaw Stock Exchange is the largest market in Central and Eastern Europe, it is underrepresented in the algorithmic trading literature. Studies of the WSE have focused on the interdependence between the Polish and other stock markets and its macroeconomic drivers [16], on tests of the informational efficiency of the exchange [17], and on the interplay between algorithmic trading, liquidity, and volatility for WSE blue-chip stocks [18]. To the best of our knowledge, multi-agent decision support systems have not been evaluated on WIG20 under an explicit walk-forward protocol with transaction costs and a risk-adjusted comparison. This gap is the primary empirical niche of the present paper; to test whether the resulting profile is specific to WSE conditions, we additionally replicate the full pipeline on the FTSE 100 under UK-specific transaction costs (Section VI-O).

D. RISK-ADJUSTED PERFORMANCE EVALUATION

A separate literature has developed a rich set of risk-adjusted metrics that go beyond the Sharpe ratio [38]. The Sortino ratio [39] penalizes only downside deviation and is therefore closer to how investors experience losses, and the Calmar ratio [40] relates annualized return to maximum drawdown, which captures the worst realized loss along the equity path. Drawdown-based measures have been extended by the Ulcer Index and the Martin ratio [41], which aggregate the full history of drawdowns rather than only the peak-to-trough extremum. The adaptive markets hypothesis of [42] argues that market efficiency is itself time-varying, which provides a theoretical motivation for evaluating a strategy across different market regimes rather than only on aggregate metrics. In line with this view, we evaluate the ensemble both on standard risk-adjusted ratios and across distinct market regimes, and we compare its equity curve against the benchmark after matching the two on ex-ante volatility. This risk-aware evaluation protocol is the third methodological contribution of the paper.

III. THEORETICAL BACKGROUND

A. FORECAST COMBINATION AND VARIANCE REDUCTION

The design of the decision agent is motivated by the theory of forecast combination. Let $f_1, \ldots, f_K$ denote K forecasts of an unobserved target y. If each f_k has forecast error $e_k = y - f_k$ with variance σ_k^2 and pairwise correlations ρ_{kj}, then for any convex combination with weights $w_k \geq 0$, $\sum_k w_k = 1$, the variance of the combined error

$$\mathrm{Var}\left(y - \sum_{k=1}^{K} w_k f_k \right) = \sum_{k=1}^{K} \sum_{j=1}^{K} w_k w_j \rho_{kj} \sigma_k \sigma_j \qquad (1)$$

is generally smaller than the variance of each individual forecast whenever the forecasts are not perfectly correlated [19]–[21]. In the special case of uncorrelated forecasts, the optimal weights are proportional to the inverse variances, $w_k \propto 1/\sigma_k^2$. Extensive empirical work has shown that simple combinations, even equal-weight averages, tend to outperform individual forecasts in a wide variety of settings [20], [21].

Two qualitative implications of (1) are particularly relevant for the design of a multi-agent trading system. First, negative pairwise correlations reduce the combined variance below what would be achievable with only non-negative correlations, which motivates the explicit inclusion of methodologically complementary agents (for example, a momentum-family agent and a mean-reversion-family agent, which are expected to produce negatively correlated signals in trendless markets). Second, the variance reduction persists as long as at least two imperfectly correlated forecasters remain in the pool, which motivates the leave-one-agent-out ablation reported in Section VI-I.

B. MAPPING TO THE TRADING PROBLEM

We map the trading problem onto this framework as follows. At a given date t and symbol s, each signal agent a produces a directional signal $\sigma_{a,s}(t) \in \{-1, 0, +1\}$ together with a confidence value $c_{a,s}(t) \in [0, 1]$. We ~~interpret~~ read $c_{a,s}(t)$ as a heuristic, monotone, bounded proxy for the inverse variance of the agent as a forecaster of the next-period directional move: higher confidence means the agent~~believes its~~'s indicator is further inside its active region, which the design treats as a sign that the forecast is more ~~precise~~ reliable at that moment. The decision agent aggregates these forecasts into a combined score

$$\mathrm{score}(t, s) \;=\; \sum_{a} w_{\kappa(a)}\, c_{a,s}(t)\, \sigma_{a,s}(t), \qquad (2)$$

where $\kappa(a) \in \{\text{trend, momentum, mean-reversion, volatility}\}$ is the methodological family of agent a and $w_{\kappa(a)}$ is a family-level weight that plays the role of a prior on the relative reliability of that family. The family weights and the decision-agent thresholds are defined in Section IV. The score is then thresholded to produce a buy, sell, or hold decision per symbol.

Two caveats define the status of this mapping and should be kept in mind throughout the paper. First, the signals and confidence values are heuristic transformations of price history, not statistically calibrated forecasts: no agent estimates a predictive distribution, and no confidence value is fitted to data. The connection to classical forecast-combination theory is therefore *conceptual* rather than literal: (1) motivates the architecture (methodologically heterogeneous agents whose errors are imperfectly correlated) and the linear form of the score (2), but it does not by itself guarantee that $c_{a,s}(t)$ behaves as an inverse-variance estimate. Second, whether higher confidence in fact identifies more accurate signals is an empirical question, which Section VI-H tests directly by relating confidence values to realized directional accuracy on the training window. The headline hypotheses below are deliberately formulated so that they do not depend on the calibration of the confidence heuristic: they concern the risk-adjusted performance of the pooled ensemble, which (1) predicts from error diversity alone.

C. HYPOTHESES

This mapping motivates three testable hypotheses. H1 frames the ensemble as a *risk-aware* decision support tool, consistent with the prediction of forecast-combination theory that combining imperfect forecasters reduces variance rather than systematically raises the mean.

- **H1.** On held-out data, net of transaction costs, the multi-agent system matches or exceeds buy-and-hold WIG20 on risk-adjusted metrics (Sharpe, Sortino, Calmar, Ulcer, Martin), on maximum drawdown, and on a volatility-matched comparison of equity curves, even if it does not dominate on gross total return.
- **H2.** On held-out data, net of transaction costs, the multi-agent system outperforms the best single-agent baseline on total return. Rationale: the combined score diversifies across agents with non-identical errors, so it should dominate any single agent except in the degenerate case where one agent is uniformly best.
- **H3.** The performance of the ensemble is robust to the removal of any single signal agent. Rationale: no single agent carries the whole directional signal, and forecast-combination theory predicts that variance reduction persists as long as at least two imperfectly correlated forecasters remain.

H1 is stated against buy-and-hold WIG20, because that is the primary panel; Section VI-O replicates the same risk-adjusted comparisons on the FTSE 100 as supplementary external evidence. H2 and H3 are evaluated on WIG20 only.

Section VI reports the evidence for each hypothesis, and Section VII discusses the degree to which the evidence is consistent with the theoretical predictions.

IV. SYSTEM DESIGN

A. ARCHITECTURE

The system is organized into four layers, as shown in Fig. 1. The *data layer* loads daily closing prices for the WIG20 constituents into a common panel format and handles missing values by dropping incomplete rows. The *signal layer* hosts the five independent signal agents. Each agent receives the history of a single symbol up to and including the current date and returns a triple: a directional signal in $\{-1, 0, +1\}$, a confidence value in $[0, 1]$, and a methodological family (trend, momentum, mean-reversion, or volatility). The *decision layer* is a single decision agent that aggregates all per-symbol signals received on a given date and produces a buy, sell, or hold recommendation per symbol. The *presentation layer* contains a portfolio simulator, which applies the recommendations at the daily closing price, maintains cash and positions net of commissions and slippage, and records the equity curve; and a metrics calculator, which computes the risk-adjusted metrics reported in Section V-E.

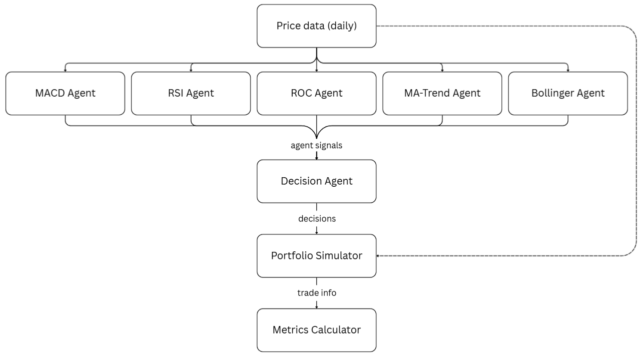

FIGURE 1. System architecture. The four layers communicate through well-defined interfaces, so that new signal agents can be added without changing the decision layer.

All signal agents inherit from a common base class that guarantees a uniform interface. Following the principle of methodological consistency, each agent uses a single family of indicators with no mixed rules. This makes the aggregation argument in Section III well-defined: each agent corresponds to a distinct forecaster f_k with its own bias-variance profile.

B. SIGNAL AGENTS

Each signal agent acts as an independent and imperfect forecaster of the next-period directional move for a single symbol. To make the forecast-combination argument of Section III well defined, every agent is confined to a single methodological family, so that its forecast errors are as distinct as possible from those of the other agents. For each agent we therefore state both its computation and its forecasting role, that is, the market behaviour it is designed to detect and the conditions under which its forecast tends to be wrong. In all cases, the agent returns a neutral (hold) signal when the history is too short to compute the indicator, and the confidence value is zero outside the active signal band.

MACD (moving-average convergence-divergence). The MACD line is the difference between a fast and a slow exponential moving average (EMA) of the closing price (default spans 8 and 17), and the signal line is an EMA of the MACD line (default span 9). The agent emits $+1$ on an upward crossover of the two lines and -1 on a downward crossover, with confidence

$$\text{confidence} = \tanh\left(\frac{|h|}{2\sigma}\right), \tag{3}$$

where h is the current histogram value and σ its standard deviation over the last 50 bars. Role: as a momentum forecaster, the agent reports the persistence of recent directional moves; its forecast errors concentrate around trend reversals, where the crossover lags the turning point. Methodological family: momentum.

RSI (relative strength index). The agent computes the average gain and the average loss over a rolling window (default 14 days), and defines $\mathrm{RSI} = 100 - 100/(1 + \mathrm{RS})$ with $\mathrm{RS} = \text{avg\_gain}/\text{avg\_loss}$. The action rule is

$$\text{action} = \begin{cases} \text{buy}, & \mathrm{RSI} < 30, \\ \text{sell}, & \mathrm{RSI} > 70, \\ \text{hold}, & \text{otherwise}, \end{cases} \tag{4}$$

with confidence equal to the normalized distance of the RSI from the nearest threshold. Role: as a mean-reversion forecaster, the agent bets against recent price extremes; its forecast errors are largest in strong trends, where the index can remain overbought or oversold for extended periods. Methodological family: mean-reversion.

ROC (rate of change). The rate of change is the percentage change in the closing price over a fixed number of days (default 10). The action rule is

$$\text{action} = \begin{cases} \text{buy}, & \mathrm{ROC} > \tau, \\ \text{sell}, & \mathrm{ROC} < -\tau, \\ \text{hold}, & \text{otherwise}, \end{cases} \tag{5}$$

where τ is a configurable threshold (default 0.02). Confidence equals $|\mathrm{ROC}|$, clipped to $[0, 1]$. Role: as a momentum forecaster on a shorter horizon than MACD, the agent responds to the raw speed of price change; its forecast errors arise when a fast move reverses abruptly. Methodological family: momentum.

Bollinger Bands. The agent computes a simple moving average (SMA) of the closing price over a window (default 20) as the middle band, and upper and lower bands $U = \text{middle} + m\sigma$, $L = \text{middle} - m\sigma$, where m is a configurable multiplier (default 2). The action rule is

$$\text{action} = \begin{cases} \text{buy}, & \text{close} < L, \\ \text{sell}, & \text{close} > U, \\ \text{hold}, & \text{otherwise}. \end{cases} \tag{6}$$

Confidence is the normalized distance of the close from the nearest band. Role: as a volatility forecaster, the agent measures how far the price has stretched from its mean in units of the standard deviation and signals a reversion when the close pierces a band; its forecast errors occur when a band breakout develops into a sustained move instead of reverting. Because the bands respond to dispersion rather than to direction, the agent carries the least directional information of the five, which is reflected in its family weight (Section IV-C). Methodological family: volatility.

MA trend. The agent uses two SMAs: fast (default 50) and slow (default 200). A buy signal is emitted when the slow SMA exceeds the fast SMA and a sell signal is emitted in the opposite case. Confidence equals the absolute difference between the two SMAs divided by the latest close. Role: as a long-horizon trend forecaster, the agent identifies the prevailing market regime from the relation between a short and a long average; its forecast errors occur in choppy, trendless markets where the two averages cross repeatedly. Methodological family: trend.

C. DECISION AGENT

The decision agent receives the list of signals produced on a given date and groups them by symbol. Signals with confidence below a global minimum confidence $c_{\min}$ are discarded. The remaining signals are aggregated into the score defined in (2). The family weights w_κ are fixed at

$$\begin{aligned} w_{\text{trend}} &= 1.3, & w_{\text{momentum}} &= 1.0, \\ w_{\text{mean-rev.}} &= 0.8, & w_{\text{volatility}} &= 0.5, \end{aligned} \tag{7}$$

The weights encode a fixed prior on the relative directional reliability of each methodological family, and their ordering follows directly from the empirical literature rather than from any in-sample search. Trend-following receives the largest weight because time-series momentum is among the most robust and persistent return predictors documented across equity markets and horizons [24], [25], and daily index constituents such as the WIG20 blue chips tend to exhibit sustained directional regimes. Momentum on a shorter horizon receives a unit weight as the reference family. Mean reversion is weighted below momentum because overbought and oversold rules are informative mainly in range-bound phases and are frequently wrong during strong trends. Volatility receives the smallest weight because band-based signals respond to dispersion rather than to direction and therefore carry the least directional information of the four families. The four levels are intentionally coarse, equally spaced multiples that keep the weights interpretable and the degrees of freedom low; they are held fixed across every symbol, regime, and evaluation window, and are never tuned on the test data. ~~The robustness of~~ Retaining a fixed prior instead of estimating the weights is additionally supported by the forecast-combination literature, in which simple fixed weights are notoriously difficult to beat with estimated ones because weight-estimation error tends to consume the gains it promises - the *forecast combination puzzle* [43], [44]. The sensitivity of the results to this fixed scheme is examined ~~by~~ empirically in Section VI-J, where each family weight is perturbed by up to $\pm 50\%$ and the whole vector is replaced by equal, inverted, in-sample-optimized, and inverse-variance-estimated alternatives; the leave-one-agent-out ablation in Section VI-I ~~, and the possibility of replacing it with adaptive or learned weights is discussed~~ covers the limiting case of removing an agent entirely, and regime-conditional weighting is discussed as future work in Section VIII. The score is then compared against mode-

TABLE 1. Taxonomy of the five signal agents. Horizon refers to the nominal lookback of the underlying indicator.

Agent	Family	Signal type	Horizon
MACD	momentum	crossover	17 d
RSI	mean-reversion	threshold	14 d
ROC	momentum	threshold	10 d
Bollinger	volatility	threshold	20 d
MA trend	trend	crossover	200 d

dependent thresholds to produce the final decision:

$$\text{action} = \begin{cases} \text{BUY}, & \text{score} \geq \theta_{\text{buy}}, \\ \text{SELL}, & \text{score} \leq \theta_{\text{sell}}, \\ \text{HOLD}, & \text{otherwise}. \end{cases} \tag{8}$$

The thresholds are $\theta_{\text{buy}} = -\theta_{\text{sell}} = 1.2$ in *passive* mode, 0.7 in *balanced* mode, and 0.3 in *aggressive* mode. In *passive* mode the decision agent additionally requires the sign of the score to agree with the dominant trend direction, which is defined as the sign-weighted sum of confidences of the trend-family agents.

The decision-agent parameters are the mode $m \in \{\text{passive}, \text{balanced}, \text{aggressive}\}$ and the minimum confidence threshold $c_{\min}$. These are the only parameters tuned during walk-forward evaluation. All indicator hyperparameters remain fixed at their literature-conventional defaults throughout.

D. AGENT TAXONOMY

Table 1 summarizes the five signal agents along three dimensions: methodological family, signal type (crossover versus threshold), and the horizon of the underlying indicator; the forecasting rationale of each family is given with the agent descriptions above. The taxonomy makes the heterogeneity of the ensemble explicit and motivates the family-level weights w_κ: agents that are expected to produce the most distinct forecast errors are pooled across families with different weights, consistent with the variance-reduction argument of Section III. The empirical correlations of the agent signals, reported in Section VI-G, support the taxonomy: cross-family pairs (for example, RSI versus ROC, or Bollinger versus ROC) exhibit the largest absolute correlations with opposing signs, consistent with their opposite interpretations of the same raw price data.

V. EXPERIMENTAL SETUP

A. DATASET

Both experiments use daily OHLCV panels from a public repository, Stooq.[1] On each market, the ensemble, the buy-and-hold index benchmark, and the LSTM directional baseline trade the same fixed roster of constituents; incomplete rows are dropped per symbol. The primary panel covers the WIG20 index and nineteen WIG20 stocks on the Warsaw Stock Exchange, excluding Żabka Group SA. whose listing on 2024/10/17 is much more recent than that of the remaining constituents. The period covers 2021/05/26 to 2026/03/31 (roughly 1,220 trading days). Values are denominated in PLN. External validation uses 20 liquid FTSE 100 constituents and the FTSE 100 index on the London Stock Exchange over the same calendar window in GBP. The anchored split is identical on both markets (train through 2024/06/30, test from 2024/07/01 through 2026/03/31); agents, hyperparameters, and evaluation procedures are unchanged, and only the price panels, index series, and transaction-cost calibration differ (Sections V-B and VI-O).

[1] https://stooq.com

B. TRANSACTION-COST MODEL

All backtests include a proportional commission of 39 basis points (bps) of notional per trade and a one-way slippage of 5 bps applied to the executed price. These values are in the typical range of Polish retail brokerages for WSE-listed equities. The commission is deducted from cash both on entry and on exit. Slippage raises the effective buy price and lowers the effective sell price by the quoted amount. Realized profit or loss is reported net of the commissions on both legs and of the round-trip slippage. This choice is conservative for a retail investor and significantly reduces the headline gross return of high-turnover configurations, which is one of the methodological concerns raised in the literature on backtest reliability [12]. A full sensitivity sweep over commission and slippage is reported in Section VI-F.

C. WALK-FORWARD PROTOCOL

The evaluation follows an anchored walk-forward protocol. The full period is split into a training window that spans 2021/05/26 to 2024/06/30 and a held-out test window that spans 2024/07/01 to 2026/03/31. On the training window we perform a grid search over the decision-agent parameters

$$\begin{aligned} \text{mode} &\in \{\text{passive}, \text{balanced}, \text{aggressive}\}, \\ c_{\min} &\in \{0, 0.05, 0.10, \ldots, 0.30\}, \end{aligned} \tag{9}$$

and select the configuration that maximizes the in-sample total return. The selected configuration is then evaluated untouched on the test window. The indicator hyperparameters are not part of the grid and remain fixed throughout. This design keeps the number of degrees of freedom low and matches the way technical-analysis rules are treated in most of the cited literature.

Total return is deliberately retained as the selection objective even though the paper evaluates the strategy primarily on risk-adjusted metrics. The reason is statistical rather than conceptual. On a low-dimensional grid of 21 configurations, in-sample total return is a simple, low-variance ranking criterion, whereas ratio statistics such as the Sharpe ratio carry substantial estimation error on samples of this length [45] and become degenerate for configurations that trade rarely: the *passive* mode produces no trades at any threshold (Table 2), so its in-sample Sharpe ratio is undefined rather than merely low, and a risk-adjusted objective would

Algorithm 1 Anchored walk-forward with precomputed signals

Input: prices P; agent set $\mathcal{A}$; modes $\mathcal{M}$; thresholds $\mathcal{C}$; train/test split $(T_{\text{tr}}, T_{\text{te}})$; costs (γ, s).
Output: selected $(m^\star, c^\star)$; test-window metrics.
$S \leftarrow \texttt{PrecomputeSignals}(P, \mathcal{A})$ {all dates, all symbols}
best $\leftarrow -\infty$
for all $m \in \mathcal{M}, c \in \mathcal{C}$ **do**
  $E_{\text{tr}} \leftarrow \texttt{Backtest}(S, T_{\text{tr}}, m, c, \gamma, s)$
  $r \leftarrow \texttt{TotalReturn}(E_{\text{tr}})$
  if $r >$ best **then**
    best $\leftarrow r$; $(m^\star, c^\star) \leftarrow (m, c)$
  end if
end for
$E_{\text{te}} \leftarrow \texttt{Backtest}(S, T_{\text{te}}, m^\star, c^\star, \gamma, s)$
return $(m^\star, c^\star), \texttt{Metrics}(E_{\text{te}})$

have to break such ties arbitrarily. To verify that this choice is not load-bearing, Section VI-N re-runs the selection under Sharpe-ratio and Calmar-ratio objectives on the identical grid and shows that the selected configuration - and therefore every out-of-sample number in the paper - is unchanged on both markets.

As a robustness check, we additionally report the results of an anchored, expanding walk-forward with three folds. For each fold $k \in \{1, 2, 3\}$, the test slice is one of three roughly equal chronological slices of the held-out window, and the training window spans $2021/05/26$ up to the day before the test slice begins. A fresh grid search is performed on each fold and the winning configuration is evaluated on the test slice of that fold. The objective of this exercise is to show that the headline configuration is not an artifact of a particular single split.

Algorithm 1 summarizes the full protocol in pseudocode. The key feature is that signals are precomputed once outside all grid-search and ablation loops; this decouples the decision-agent grid search from signal computation, reduces the end-to-end runtime by more than an order of magnitude, and makes the cost-sensitivity sweep of Section V-H cheap because changing costs does not require recomputing any agent signal.

D. BASELINES

Three families of baselines are used. The first is *index buy-and-hold*: on WIG20 the benchmark invests the initial cash into the WIG20 index on the first date of the test window and holds the position until the end; on the FTSE replication the same rule applies to the FTSE 100 index. Buy-and-hold is a standard benchmark in the evaluation of trading strategies [14]. The second family is a set of *single-agent* runs in which the decision agent receives signals from only one signal agent; this produces five baselines, one per indicator. In addition, we run five *leave-one-agent-out* configurations, each of which removes one agent from the ensemble. The third family ~~is a~~ comprises two ~~deep-learning baseline~~machine-learning baselines, described next. All baselines are evaluated on the same test window on each market, use market-specific transaction costs described in Sections V-B and VI-O, and adopt the same decision-agent configuration selected by the walk-forward protocol. The only exceptions are buy-and-hold, which has no decision-agent parameters, and the ~~deep-learning baseline, which replaces~~ machine-learning baselines, which replace the decision agent with a learned classifier.

1) ~~Deep-Learning Baseline~~Machine-Learning Baselines

As a modern learning-based reference point, we implement an LSTM directional classifier in the style of [4]. For each symbol the model consumes a rolling window of causal per-day features derived from the same price history available to the agents (multi-horizon returns, realized volatility, and the normalized RSI, ROC, MACD-histogram, Bollinger %b, and moving-average-spread indicators), and predicts the probability that the h-day-ahead return is positive. The network is a single-layer LSTM followed by dropout and a linear output unit, trained with the Adam optimizer under binary cross-entropy. To keep the comparison fair, all model-selection choices are made strictly in-sample: the prediction horizon h, the operating threshold, and early stopping are selected on an internal validation slice carved chronologically from the training window, and the held-out test window is never used for any tuning. Feature standardization statistics are estimated on the training window only, so the procedure carries no look-ahead. Per-symbol probabilities are mapped to long or flat positions through a symmetric no-trade band around 0.5: a probability above $0.5 + \delta$ opens or holds a long position, a probability below $0.5 - \delta$ closes it, and intermediate probabilities leave the current position unchanged, which mirrors the hold behavior of the decision agent and prevents the excessive turnover that would otherwise be induced by daily re-prediction. On both markets the in-sample selection returns a horizon of $h = 3$ days and a band of $\delta = 0.03$. The resulting long/flat signals are executed through the same portfolio simulator, under identical costs and position sizing, so that any difference reflects the decision rule rather than the accounting.

The second machine-learning baseline is a gradient-boosted-trees (GBM) directional classifier [29], implemented with LightGBM [30]. Gradient-boosted trees are consistently among the strongest learners on tabular financial features in large-scale empirical comparisons [31], [32], which makes them a natural, structurally different complement to the recurrent LSTM. For each symbol and day the model consumes the same causal per-day features as the LSTM, augmented with lagged copies of each feature at lags of up to 20 trading days so that the flat feature vector carries a temporal context comparable to the LSTM lookback window, and predicts the probability that the h-day-ahead return is positive. Training pools feature rows across symbols, splits them chronologically into training and validation slices, applies early stopping on validation log-loss, and averages three differently seeded models. Exactly as for the LSTM, the prediction horizon and the no-trade band are selected strictly in-sample by the net-of-cost Sharpe ratio of a validation-slice portfolio simulation, and the resulting long/flat signals are executed through the same portfolio simulator under identical costs. The in-sample selection returns $h = 1$ and $\delta = 0.03$ on WIG20, and $h = 3$ and $\delta = 0.03$ on the FTSE 100.

E. RISK-ADJUSTED METRICS

The following metrics are computed on the equity curve and on the list of closed trades. **Total return** is the percentage change in portfolio value from the first to the last date of the evaluation window. **CAGR** (compound annual growth rate) is the annualized return over the same window. **Max drawdown** is the largest peak-to-trough decline, reported as a negative fraction. **Sharpe ratio** [38] is the annualized ratio of excess return over standard deviation, using a factor of $\sqrt{252}$ and a risk-free rate of zero. **Sortino ratio** [39] replaces the denominator with downside deviation, which penalizes only below-target returns:

$$\text{Sortino} = \frac{\sqrt{252}\,\bar{r}}{\sqrt{\frac{1}{N}\sum_{t=1}^{N}\min(r_t, 0)^2}}, \tag{10}$$

where $\bar{r}$ is the mean daily return and the denominator is the downside deviation. **Calmar ratio** [40] is CAGR divided by the absolute value of the maximum drawdown. **Ulcer Index** [41] aggregates the full drawdown path:

$$\text{UI} = 100 \cdot \sqrt{\frac{1}{N}\sum_{t=1}^{N}\text{dd}_t^{\,2}}, \tag{11}$$

where dd_t is the running drawdown of the equity curve at date t. **Martin ratio** [41] is the annualized excess return divided by the Ulcer Index. The Ulcer and Martin statistics are more sensitive to prolonged drawdowns than the peak-to-trough maximum, and we use them as tie-breakers whenever two strategies have similar peak drawdowns but different drawdown profiles. **Number of trades** is the count of closed trades and **Win rate** is the fraction of closed trades with positive realized profit.

F. MARKET-REGIME LABELING

To avoid comparing the ensemble and buy-and-hold only on pooled aggregates, we label each trading day of the test window as belonging to one of three regimes: *bull*, *correction*, or *sideways*. The labeling uses only information available up to date t from the WIG20 index itself, so it does not peek at future prices. Let high_t be the rolling 252-day maximum of the WIG20 closing price at t, and let ret_t be the price relative $\text{close}_t/\text{high}_t$. A day is labeled *correction* when $\text{ret}_t < 1 - 0.10$, that is, the index is at least ten percent below its trailing one-year high. A day is labeled *bull* when $\text{ret}_t \geq 1 - 0.05$, that is, within five percent of the trailing high, and the 50-day SMA exceeds the 200-day SMA of the WIG20 index. All other days are labeled *sideways*. These thresholds are standard in the practitioner literature on market regimes and are not tuned on the training data. Applying the labelling to the test window produces 52.87% bull days, 21.38% correction days, and 25.75% sideways days (Table 6), which means the test window covers all three regimes and the ensemble is evaluated under each of them.

G. VOLATILITY-MATCHED COMPARISON

The volatility-matched comparison is an analytical normalisation, not a directly implementable trading result; its purpose is to place the ensemble and the benchmark on the same risk budget so that their returns can be compared on equal terms. The intuition is straightforward. Because the ensemble is out of the market on many days, it runs at a lower volatility than the fully invested benchmark, and a raw comparison of cumulative returns therefore penalises the ensemble for taking less risk. To remove this confound, the ensemble returns are multiplied by a single constant chosen so that the strategy and the benchmark have the same ex-ante volatility, after which the two equity curves are compared.

The formal construction is as follows. A standard criticism of the Sharpe ratio is that it is scale-invariant: it evaluates return per unit risk, but not the absolute level of risk [13]. A strategy with a Sharpe ratio of 1.0 and a realized annualized volatility of 10% produces a smaller cumulative return than a strategy with a Sharpe ratio of 1.0 and a realized volatility of 20%, simply because the second strategy takes more risk. To remove this confound, we report a volatility-matched comparison in which the equity curve of the ensemble is scaled, ex ante, to the realized volatility of buy-and-hold WIG20 on the training window. The target volatility is the annualized standard deviation of WIG20 daily returns over the training window, $\sigma^{\text{target}} = \sqrt{252}\,\text{std}(r_{\text{train}}^{\text{WIG20}}) = 0.22$. The strategy volatility on the training window is $\sigma^{\text{strat}} = \sqrt{252}\,\text{std}(r_{\text{train}}^{\text{ensemble}}) = 0.11$. The leverage factor applied to the out-of-sample ensemble return is $\lambda = \sigma^{\text{target}}/\sigma^{\text{strat}} = 2.06$, constant over the test window. Because λ is computed purely from training-window data, the volatility-matched comparison is genuinely ex ante and does not peek at out-of-sample returns. The scaled equity curve and its metrics are reported in Section VI-D. Two cautions apply to this comparison. First, the unleveraged ensemble remains the primary finding of the paper; the volatility-matched figures are a risk-budget counterfactual, not a recommendation to trade with leverage. Second, the required leverage factor of $\lambda = 2.06$ is above the level that most retail investors on the WSE can realistically access or would prudently use, so the volatility-matched results should be read as an analytical upper bound on the risk-normalised return of the strategy rather than as an attainable outcome.

H. COST-SENSITIVITY SWEEP

In addition to the baseline calibration of 39 bps commission and 5 bps slippage, we sweep the transaction-cost parameters over a grid

$$\begin{aligned} \text{commission} &\in \{0, 10, 20, 39, 60, 80, 100\} \text{ bps}, \\ \text{slippage} &\in \{0, 5, 10, 20\} \text{ bps}, \end{aligned} \tag{12}$$

and re-run the walk-forward backtest at each grid point. Crucially, the agent signals are precomputed once and passed into each backtest, so that the cost sweep changes only the portfolio-level accounting. The purpose of this experiment is to quantify how much of the risk-adjusted edge of the ensemble over buy-and-hold comes from the specific cost calibration. Section VI-F reports the full grid.

I. BOOTSTRAP CONFIDENCE INTERVALS AND TESTS

In intuitive terms, a bootstrap repeatedly redraws the observed daily returns at random with replacement, recomputes the metric of interest on each redrawn series, and uses the spread of the resulting values to quantify how much the metric could have varied by chance. This procedure makes no assumption that returns follow a normal distribution. Point estimates of total return and Sharpe ratio are accompanied by a 95% percentile bootstrap confidence interval. The bootstrap is computed on the vector of daily returns of the equity curve, with $10{,}000$ resamples drawn with replacement. For the comparison against the local index buy-and-hold benchmark, against the best single-agent baseline, and against the LSTM directional baseline of Section V-D1, a paired bootstrap test is performed: at each resample, the same set of dates is used for both series, the difference of the statistic of interest is computed, and the one-sided p-value reports the fraction of resamples with a non-positive difference. We treat $p < 0.05$ as evidence that the strategy dominates the baseline on the test window. The bootstrap is a natural choice here because the distribution of trading-strategy metrics is known to be non-Gaussian and path-dependent, and bootstrap methods have been advocated for this setting in prior work [15]. As is standard in the literature on trading-strategy evaluation, we do not expect the bootstrap tests to be powered to detect moderate effect sizes on a single test window of just under two years [12]; we therefore report the p-values alongside point estimates and confidence intervals rather than relying on them as the sole arbiter of performance.

VI. RESULTS

A. PARAMETER SELECTION ON TRAINING DATA

Table 2 reports the in-sample total return of every combination of decision-agent mode and minimum confidence on the training window. The *passive* mode rarely reaches its high buy/sell thresholds, and for every value of minimum confidence it produces no trades. The *balanced* mode trades moderately, with positive but small total return across all thresholds. The *aggressive* mode produces the highest total returns overall, and the best configuration on the training window is (*aggressive*, $c_{\min} = 0.05$), which is the configuration selected for out-of-sample evaluation. The result is consistent with the well-known observation that moderate signal tolerance on a trend-rich daily dataset tends to maximize gross return, although net return is reduced noticeably by the transaction-cost model of Section V-B.

TABLE 2. In-sample total return (train window, 2021-2024) by decision-agent mode and minimum confidence. Higher is better. Selected configuration in bold.

$c_{\min}$	passive	balanced	aggressive
0.00	0.00	0.05	0.48
0.05	0.00	0.09	**0.55**
0.10	0.00	0.09	0.49
0.15	0.00	0.11	0.40
0.20	0.00	0.23	0.28
0.25	0.00	0.16	0.25
0.30	0.00	0.28	0.23

TABLE 3. Out-of-sample metrics on the held-out test window, net of transaction costs. Brackets give 95% bootstrap confidence intervals.

	Multi-agent system	Buy-and-hold WIG20
Total return	0.23 [−0.10, 0.68]	0.31
CAGR	0.13	0.17
Max drawdown	−0.06	−0.17
Sharpe ratio	1.06 [−0.43, 2.57]	0.82
Calmar ratio	2.10	0.96
Number of trades	34	–
Win rate	0.65	–

B. HEADLINE RESULTS ON HELD-OUT DATA

Table 3 reports the headline metrics on the held-out test window for the selected configuration and for buy-and-hold WIG20. All numbers are net of the transaction costs described in Section V-B. On gross total return, buy-and-hold leads: it earns 0.31 over 435 trading days, whereas the ensemble earns 0.23 (CI $[-0.10, 0.68]$), and the one-sided bootstrap p-value for the hypothesis that the ensemble exceeds buy-and-hold on total return is 0.636. Once the comparison moves to the risk axis, the picture reverses. The Sharpe ratio of the ensemble is 1.06 (CI $[-0.43, 2.57]$), compared with 0.82 for buy-and-hold, and the p-value on Sharpe is 0.297. The maximum drawdown is -0.06 against -0.17, a reduction of more than a factor of two. The Calmar ratio is 2.10 against 0.96 for buy-and-hold, more than twice as large. Fig. 2 shows the two equity curves, and Fig. 3 overlays the underwater (drawdown) curves of the two series.

C. RISK-ADJUSTED COMPARISON

The reversal of the comparison on the risk axis is reinforced by the extended risk-adjusted metrics reported in Table 4. The downside deviation of the ensemble is 0.08 against 0.15 for buy-and-hold, a reduction of more than forty percent. Consequently, the Sortino ratio rises from 1.19 for buy-and-hold to 1.60 for the ensemble. The Ulcer Index, which captures the integrated depth and persistence of drawdowns rather than only their peak, is 2.39 for the ensemble against 6.69 for buy-and-hold, a reduction of nearly two-thirds. The Martin ratio rises from 2.49 to 5.29, more than a factor of two. Fig. 4 plots the rolling 60-day Sharpe of both series and shows that

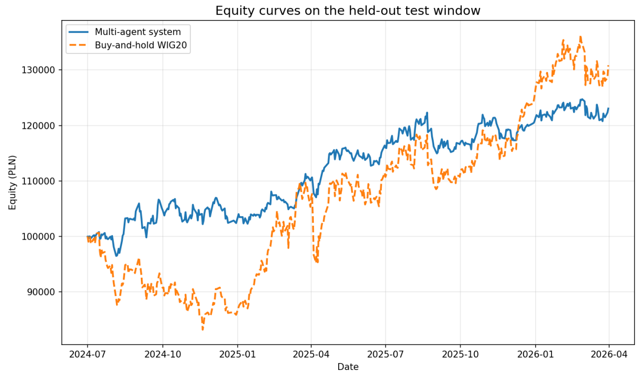

FIGURE 2. Equity curves on the held-out test window for the multi-agent system (selected configuration) and for buy-and-hold WIG20. Both series start at the same initial cash and are net of transaction costs.

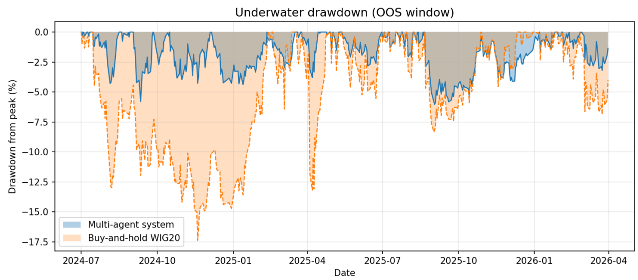

FIGURE 3. Underwater (drawdown) curves for the ensemble and for buy-and-hold WIG20 on the held-out test window. The drawdowns of the ensemble are both shallower and shorter, which is the central risk-adjusted finding of the paper.

TABLE 4. Risk-adjusted metrics on the held-out test window, net of transaction costs.

Metric	Multi-agent system	Buy-and-hold WIG20
Sharpe ratio	1.06	0.82
Sortino ratio	1.60	1.19
Downside deviation	0.08	0.15
Calmar ratio	2.10	0.96
Max drawdown	−0.06	−0.17
Ulcer Index	2.39	6.69
Martin ratio	5.29	2.49

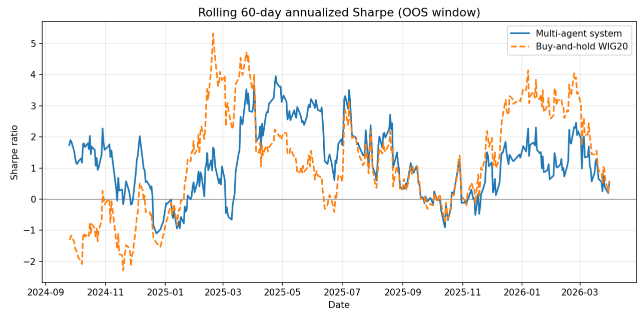

FIGURE 4. Rolling 60-day Sharpe ratio of the ensemble and of buy-and-hold WIG20 on the held-out test window. The rolling Sharpe of the ensemble exceeds buy-and-hold during correction subperiods and is close to it during the bull leg.

the rolling Sharpe of the ensemble is above the buy-and-hold rolling Sharpe for most of the test window, and in particular during the correction subperiods.

D. VOLATILITY-MATCHED COMPARISON

The headline comparison on gross total return is systematically biased against the ensemble because the two series run at very different volatilities. On the training window, the annualized volatility of the ensemble is 0.11, whereas that of buy-and-hold WIG20 is 0.22. To put the two series on the same risk budget, we scale the out-of-sample daily returns of the ensemble by the ex-ante leverage factor $\lambda = 2.06$ defined in Section V-G. Table 5 reports the resulting metrics. On the volatility-matched basis, the total return of the ensemble rises to 0.49, compared with 0.31 for buy-and-hold, a gain of $+18.55$ percentage points. The Sharpe ratio is unchanged by construction (1.06), because Sharpe is scale-invariant in the leverage factor when cash reserves are ignored; the Sortino ratio is 1.63, against 1.19; and the Calmar ratio is 2.13, against 0.96. The volatility-matched maximum drawdown is -0.12, which is still materially smaller than the buy-and-hold drawdown of -0.17 despite the much higher leverage, because the correction-regime drawdown of the ensemble is structurally shallower (Section VI-E). We emphasize that these volatility-matched figures are analytical rather than directly implementable: leverage of magnitude $\lambda = 2.06$ is impractical for a typical retail investor on the WSE, so Table 5 should be read as a risk-normalised counterfactual and not as an attainable trading outcome. Its role is to correct for the fact that comparing a partially-invested strategy at its natural volatility against a fully-invested benchmark at roughly twice that volatility understates the risk-per-unit-of-return advantage of the strategy.

E. MARKET-REGIME DECOMPOSITION

The aggregate metrics in Table 3 mix three very different market regimes within the test window. Using the labeling described in Section V-F, the test window contains 230 bull days (52.87% of the window), 93 correction days (21.38%), and 112 sideways days (25.75%). Table 6 decomposes the total return and the Sharpe ratio of both the ensemble and buy-and-hold WIG20 by regime. The ensemble loses -0.07 during the correction subperiods, whereas buy-and-hold loses -0.20: the correction-regime defense of the ensemble is $+12.57$ percentage points. During the sideways subperiods, the ensemble earns 0.05, whereas buy-and-hold earns -0.00: the sideways-regime excess of the ensemble is $+5.14$ percentage points. During the bull subperiods, the ensemble earns 0.26 against 0.63 for buy-and-hold: the bull-regime shortfall is -36.84 percentage points. The decomposition reconciles the aggregate numbers in Table 3: buy-and-hold is fully exposed to the bull leg and captures most of its upside, while the ensemble, by design, reduces exposure during the correction and sideways regimes and thereby gives up some of the bull-regime upside. In a test window that is dominated by a bull

TABLE 5. Volatility-matched out-of-sample metrics. The ensemble is scaled, ex ante, to the buy-and-hold volatility measured on the training window (leverage factor $\lambda = 2.06$).

Metric	Ensemble (unleveraged)	Ensemble (vol-matched)	Buy-and-hold WIG20
Total return	0.23	0.49	0.31
CAGR	0.13	0.26	0.17
Sharpe ratio	1.06	1.06	0.82
Sortino ratio	1.60	1.63	1.19
Max drawdown	−0.06	−0.12	−0.17
Calmar ratio	2.10	2.13	0.96
Ulcer Index	2.39	4.96	6.69

regime, the aggregate total return will mechanically favor the fully-invested benchmark; in a test window with a larger correction share, the aggregate comparison would favor the ensemble.

F. COST-SENSITIVITY ANALYSIS

Table 7 reports the out-of-sample Sharpe ratio over the cost grid defined in Section V-H. At zero costs the Sharpe is 1.15; at the baseline calibration of 39 bps commission and 5 bps slippage the Sharpe is 1.06; at the most punitive configuration of 100 bps commission and 20 bps slippage the Sharpe drops to 0.92. Over the entire grid, the minimum Sharpe of 0.92 exceeds the buy-and-hold Sharpe of 0.82. The total-return column (omitted from the table for brevity but computed along with Sharpe) shows the same qualitative pattern: the ensemble trails buy-and-hold on gross total return at every grid point, and at every grid point it achieves a higher Sharpe, higher Sortino, and smaller drawdown. This indicates that the risk-adjusted edge of the ensemble is not an artifact of the specific cost calibration; it is a stable property of the strategy over a plausible range of retail brokerage fees.

G. EMPIRICAL SIGNAL CORRELATIONS

The variance-reduction argument of Section III rests on the claim that the five agents produce signals with imperfectly correlated errors. Table 8 reports the pairwise Pearson correlations [46] of the signed confidence values produced by the agents, computed across all pooled (symbol, date) observations in the training window. The mean absolute off-diagonal correlation is 0.25 across 10 distinct pairs, with extremes of −0.71 (RSI vs. ROC) and 0.46 (RSI vs. Bollinger). In particular, the momentum family (MACD, ROC) and the mean-reversion family (RSI) exhibit negative correlations, as expected from their opposite interpretations of the same raw price input; the trend agent (MA trend) is nearly uncorrelated with all other agents, as expected from its long horizon. Fig. 5 renders the correlation matrix as a heatmap. These empirical correlations directly instantiate the non-identical-errors assumption of (1), and Section VII returns to them to explain why the ensemble achieves most of its variance reduction from the interaction between the momentum and mean-reversion families rather than from the trend agent on its own.

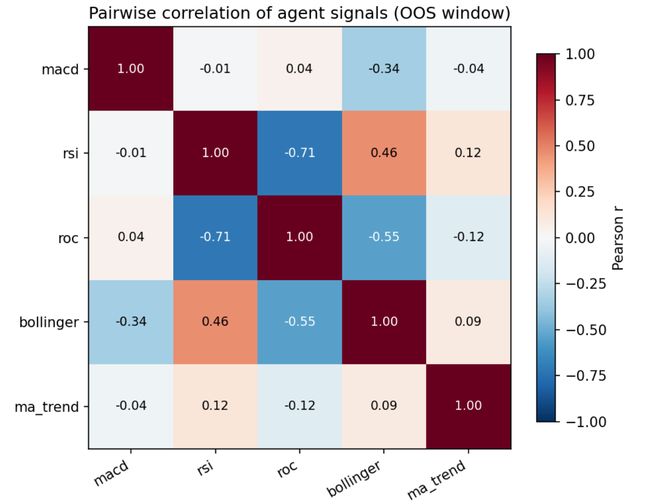

FIGURE 5. Heatmap of the pairwise Pearson correlations of the agent signed-confidence signals on the training window. Blue cells mark negative correlations, red cells mark positive ones. The largest absolute correlations appear in the RSI-ROC and RSI-Bollinger pairs, which is consistent with their shared reliance on overbought/oversold-style rules.

H. VALIDATION OF THE CONFIDENCE MEASURE

Section III motivates the score (2) by reading the per-signal confidence as a heuristic proxy for inverse forecast variance. If that reading were literally correct, higher-confidence signals should be directionally right more often than lower-confidence ones. Table 9 tests this implication on the training window, so that no test-window information is used: for every agent, all active (non-hold) signals are pooled across symbols and dates, the directional hit rate against the next-day move is computed, and the association between confidence and correctness is measured both by a Spearman rank correlation [47] and by comparing the hit rates of the bottom and top confidence quintiles, in the spirit of the reliability analysis of [48]. A rank correlation is the appropriate statistic here because correctness is binary and the confidence distributions are strongly skewed towards zero, so no linearity assumption is warranted; this complements the Pearson correlations of Section VI-G [46], which measure the association *between* agents rather than between an agent's confidence and its own accuracy.

TABLE 6. Decomposition of the held-out test window into bull, correction, and sideways subperiods. Shares refer to the fraction of trading days in the test window.

Regime	Days	Share	Ensemble TR	Ensemble Sharpe	B&H TR	B&H Sharpe
Bull	230	52.9%	0.26	2.47	0.63	2.76
Correction	93	21.4%	−0.07	−1.34	−0.20	−2.41
Sideways	112	25.7%	0.05	0.93	−0.00	0.11
All	435	100.0%	0.23	1.06	0.31	0.82

TABLE 7. Cost sensitivity of the out-of-sample Sharpe ratio. Rows are commission in basis points; columns are one-way slippage in basis points.

Commission (bps)	0 bp	5 bp	10 bp	20 bp
0	1.15	1.14	1.13	1.11
10	1.13	1.12	1.11	1.09
20	1.11	1.10	1.09	1.07
39	1.07	1.06	1.05	1.03
60	1.03	1.02	1.02	0.99
80	0.99	0.99	0.98	0.96
100	0.96	0.95	0.94	0.92

TABLE 8. Pearson correlations of the pooled (symbol, date) signed confidence values of the five signal agents, computed on the training window. Upper triangle shown.

	MACD	RSI	ROC	Bollinger	MA trend
MACD	**1.00**	-0.01	+0.04	-0.34	-0.04
RSI		**1.00**	-0.71	+0.46	+0.12
ROC			**1.00**	-0.55	-0.12
Bollinger				**1.00**	+0.09
MA trend					**1.00**

TABLE 9. Confidence validation on the WIG20 training window at the one-day horizon. ρ is the Spearman rank correlation between confidence and directional correctness, with its p-value; Hit (Q1) and Hit (Q5) are hit rates in the bottom and top confidence quintiles.

Agent	n	Hit rate	ρ	p	Hit (Q1)	Hit (Q5)
MACD	1,405	0.472	+0.002	0.955	0.470	0.484
RSI	3,847	0.497	−0.035	0.031	0.525	0.497
ROC	11,155	0.498	+0.016	0.082	0.494	0.520
Bollinger	1,674	0.483	−0.004	0.867	0.478	0.481
MA trend	10,982	0.505	−0.004	0.690	0.503	0.500
Pooled	29,063	0.498	+0.003	0.578	0.493	0.495

TABLE 10. Single-agent baselines on the test window, with the selected mode and minimum confidence. All numbers net of costs.

Agent used	Total return	Sharpe ratio
MACD only	0.15	0.86
RSI only	0.16	1.03
ROC only	0.00	0.13
Bollinger only	0.00	0.00
MA trend only	−0.00	0.03
Full ensemble	0.23	1.06

The outcome is unambiguous: the confidence values carry no usable information about signal-level accuracy. Across 29,063 pooled training-window signals the hit rate is 0.498, the per-agent rank correlations span $[-0.035, 0.016]$ with none exceeding $|\rho| = 0.035$, and the top confidence quintile is not systematically more accurate than the bottom one. The only nominally significant entry in Table 9 (RSI, $p = 0.031$) is negative - higher confidence associates with marginally *worse* calls - explains well under one percent of the variance in correctness, and does not survive a multiple-testing correction across the ten agent-horizon combinations tested, so it argues against rather than for the inverse-variance reading. The picture is unchanged at a five-day horizon (pooled hit rate 0.510), and the FTSE 100 training window reproduces it (largest per-agent $|\rho| = 0.027$, pooled hit rate 0.502). Two consequences follow. First, the inverse-variance reading of the confidence values is a conceptual analogy, not an empirically calibrated property: at the level of individual signals the agents are close to uninformed forecasters at short horizons, and the magnitude of their confidence does not separate good calls from bad ones. This is consistent with the low tuned threshold ($c_{\min} = 0.05$), which in practice acts as a mild noise filter rather than as a precision weight. Second, the risk-adjusted performance documented in the preceding subsections cannot be attributed to precision-weighted averaging; the operative mechanism is the pooling of weakly and negatively correlated signals across methodological families (Table 8), which lowers the variance of the aggregate position irrespective of whether the individual confidence magnitudes are informative, together with the regime exposure profile that the thresholded score induces (Table 6). Section VII returns to the implications of this finding for the theoretical framing of the paper.

I. SINGLE-AGENT BASELINES AND LEAVE-ONE-OUT ABLATION

Table 10 reports the test-window total return and Sharpe ratio of the five single-agent baselines and of the full ensemble. The best single-agent is `only_rsi` with a total return of 0.16. The one-sided bootstrap p-value for the hypothesis that the ensemble beats the best single-agent baseline on total return is 0.268. These numbers are the primary evidence for H2.

Table 11 reports the effect of removing each signal agent from the ensemble. No single agent can be removed without some effect on the headline metrics, but in all cases the ensemble without that agent still leads buy-and-hold WIG20 on Sharpe ratio and maximum drawdown on point estimates. This is the primary evidence for H3.

TABLE 11. Leave-one-agent-out ablation on the test window. The variant column indicates which agent was removed.

Variant	Total return	Sharpe ratio
Full ensemble	0.23	1.06
no MACD	0.28	1.28
no RSI	0.23	0.92
no ROC	0.26	1.17
no Bollinger	0.23	1.06
no MA trend	0.19	1.16

TABLE 12. Alternative weighting schemes on the WIG20 test window, with (mode, $c_{\min}$) fixed at the walk-forward selection. Weights are listed as (trend, momentum, mean-reversion, volatility).

Scheme	Weights	TR	Sharpe	MDD
Baseline	$(1.3, 1.0, 0.8, 0.5)$	0.23	1.06	-0.06
Equal	$(1.0, 1.0, 1.0, 1.0)$	0.25	1.16	-0.06
Inverted	$(0.5, 0.8, 1.0, 1.3)$	0.23	1.23	-0.05
Optimized	$(1.30, 0.80, 0.80, 0.50)$	0.25	1.18	-0.05
Inv.-variance	$(1.02, 1.00, 1.00, 0.98)$	0.25	1.16	-0.06
Buy-and-hold	–	0.31	0.82	-0.17

J. ~~COMPARISON WITH A DEEP-LEARNING BASELINE~~WEIGHT SENSITIVITY AND ALTERNATIVE WEIGHTING SCHEMES

The family weights $w_{\mathcal{F}}$ are the only fixed prior in the decision layer, so we quantify how much the headline results depend on them. Every experiment in this subsection holds the walk-forward-selected configuration (`aggressive`, $c_{\min} = 0.05$) fixed and varies only the weights, so that any difference is attributable to the weighting scheme alone. Three families of alternatives are evaluated on the held-out window. First, each family weight is perturbed one at a time by $\pm 25\%$ and $\pm 50\%$, giving 16 perturbed configurations. Second, two fixed alternatives discard the ordering rationale of Section IV-C entirely: equal weights, and a fully inverted ordering in which volatility receives the largest weight and trend the smallest. Third, two data-driven schemes are estimated on the training window only: *optimized* weights, selected by an exhaustive grid over the four family weights under the same in-sample total-return objective as the headline protocol, and *inverse-variance* weights, proportional to the inverse of each family's directional error variance estimated from training-window signal outcomes, which operationalizes the classical prescription of [19]. Table 12 reports the named schemes on WIG20.

The results support three conclusions. First, the risk-adjusted profile is robust to the weights. Across the 16 perturbations the out-of-sample Sharpe ratio spans $[0.70, 1.38]$ and total return spans $[0.14, 0.30]$, and only 2 of the 20 fixed and theory-implied schemes fall below the buy-and-hold Sharpe ratio of 0.82; the four volatility-weight perturbations leave every metric exactly unchanged, because the Bollinger agent is essentially inactive under the selected configuration. The baseline is not the best configuration in hindsight - moderately downweighting the momentum-family agents would have improved this particular window, consistent with the ablation finding that removing MACD helps. The two schemes below the benchmark Sharpe ratio do surrender that part of the risk-adjusted advantage, but even they keep the maximum drawdown markedly shallower than the buy-and-hold -0.17, so the drawdown-reduction profile - the central risk-adjusted finding of the paper - survives every weighting scheme even where the Sharpe advantage does not. Second, in-sample optimization adds little. The grid-optimized weights land almost exactly on the fixed prior (only the momentum weight moves, from 1.0 to 0.80), and their out-of-sample total return of 0.25 is within about two percentage points of the baseline; on the FTSE 100 the optimized weights actually *underperform* the fixed prior out of sample (0.13 against 0.14 total return), a direct instance of the forecast combination puzzle, in which the estimation error of fitted weights consumes the gains they promise [43], [44]. Third, the inverse-variance prescription is uninformative here: because the per-family directional hit rates are nearly identical (Section VI-H), the estimated weights come out close to equal (1.02, 1.00, 1.00, 0.98 for trend, momentum, mean reversion, and volatility), and the resulting performance is practically identical to the equal-weight scheme. The FTSE 100 replication of the full experiment shows the same qualitative pattern, with perturbation Sharpe ratios spanning $[0.69, 1.54]$ and 3 of 20 fixed schemes below the buy-and-hold Sharpe ratio. Together with the leave-one-agent-out ablation, these results indicate that the contribution of the ensemble does not hinge on the particular weight values, and that the coarse fixed prior is defensible against perturbed, reordered, optimized, and theory-implied alternatives.

K. COMPARISON WITH MACHINE-LEARNING BASELINES

Table 13 compares the ensemble with the ~~LSTM directional baseline~~ two machine-learning baselines of Section V-D1 on the held-out WIG20 test window. The LSTM earns a total return of 0.08 at a Sharpe ratio of 0.44, against 0.23 and 1.06 for the ensemble and 0.31 and 0.82 for buy-and-hold. The ensemble exceeds the LSTM on total return, Sharpe ratio, and maximum drawdown (-0.06 against -0.11), and it does so while placing far fewer trades (34 against 237). The one-sided paired bootstrap p-values for the ensemble exceeding the LSTM are 0.068 on total return and 0.063 on Sharpe ratio; as with the buy-and-hold comparison, the point-estimate advantage is sizable but does not reach the conventional $p < 0.05$ threshold on a single test window. ~~The learned classifier, trained end-to-end on the same price history and executed under identical costs, does not match~~

The GBM baseline behaves similarly. It earns 0.04 at a Sharpe ratio of 0.47 over 34 closed trades; its equity path is statistically indistinguishable from the LSTM's (paired one-sided $p = 0.621$ on total return), and the ensemble leads it on total return, Sharpe ratio, and Calmar ratio (one-sided $p = 0.107$ on total return and $p = 0.210$ on Sharpe ratio). The single metric on which the GBM edges ahead is the

TABLE 13. Comparison of the ensemble, the LSTM and GBM directional ~~baseline~~baselines, and buy-and-hold on the held-out WIG20 test window, net of transaction costs.

Metric	Multi-agent ~~system~~	LSTM ~~baseline~~	GBM	Buy-and-hold ~~WIG20~~
Total return	0.23	0.08	0.04	0.31
Sharpe ratio	1.06	0.44	0.47	0.82
Max drawdown	−0.06	−0.11	−0.05	−0.17
Calmar ratio	2.10	0.42	0.45	0.96
Number of trades	34	237	34	–

TABLE 14. Anchored walk-forward with $K = 3$ folds. Selected mode and minimum confidence per fold, plus the test-window total return of each fold.

Fold	Selected mode	$c_{\min}$	Test total return
1	`aggressive`	0.05	0.05
2	`aggressive`	0.05	0.09
3	`aggressive`	0.10	0.08

maximum drawdown (-0.05 against -0.06), but this reflects an almost flat equity curve rather than better risk control: the model takes very little risk and earns correspondingly little, which is exactly what the Calmar ratio registers (0.45 against 2.10). The fact that two structurally different learners - a recurrent neural network and a gradient-boosted tree ensemble, each tuned strictly in-sample under the identical protocol - arrive at similarly modest out-of-sample results on this market strengthens the earlier reading: the WIG20 test window offers little exploitable signal to flexible learners, and the transparent forecast-combination ensemble ~~on this market, which is~~ remains competitive without any training at all, consistent with the broader finding that simple, well-regularized combinations are competitive with more flexible learners on noisy financial series [20], [21].

L. STATISTICAL SIGNIFICANCE

The bootstrap significance tests attach two kinds of evidence to the point estimates discussed above. First, the 95% percentile confidence intervals around the headline metrics reflect the sampling uncertainty inherent in a test window of 435 trading days. The total-return confidence interval for the ensemble spans $[-0.10, 0.68]$ and the Sharpe-ratio interval spans $[-0.43, 2.57]$. The buy-and-hold point estimates for total return (0.31) and Sharpe ratio (0.82) both lie strictly inside these intervals, which is a statistical way of saying that one cannot reject the hypothesis that the two series share the same underlying mean return and Sharpe ratio on the test window. Second, the paired bootstrap tests yield one-sided p-values of 0.636 for the total-return comparison against buy-and-hold and 0.297 for the Sharpe comparison. Neither test crosses the conventional $p < 0.05$ threshold.

We interpret these results in two ways. The test-window point estimates on the risk axis (drawdown, Sortino, Ulcer, Martin) are consistently favorable to the ensemble by margins that are economically large and reasonably stable across the ablation and cost grids, but the null hypothesis of equal mean performance is hard to reject on a single window of under two years because trading-strategy return distributions are heavy-tailed and path-dependent [12], [15]. Consistent with standard practice in the literature, we therefore report point estimates, confidence intervals, and p-values together and do not rely on p-values as a sole arbiter. Second, the fact that the buy-and-hold point estimates lie inside the bootstrap confidence intervals of the ensemble supports the interpretation that, on gross total return, the two series are statistically indistinguishable, whereas the large and consistent gap on drawdown-based and downside-based metrics supports the risk-aware interpretation of the contribution of the ensemble (Section VII).

Two points deserve emphasis for the correct reading of these results. The first concerns the width of the confidence intervals. The total-return interval $[-0.10, 0.68]$ and the Sharpe interval $[-0.43, 2.57]$ are wide because a held-out window of 435 trading days contains relatively few independent realizations of the estimated quantities, and because heavy-tailed and serially dependent returns inflate the variance of the resampled statistics. With a sample of this length the paired bootstrap test has limited power: even a genuine moderate edge would frequently fail to reach the conventional $p < 0.05$ threshold. The non-significant p-values should therefore be read as a consequence of limited statistical power on a short out-of-sample window, not as positive evidence that the two strategies are equivalent. The second point concerns the distinction between practical and statistical significance. The advantages of the ensemble on the drawdown-based and downside-based metrics are practically significant, in the sense that they are economically large and stable across the ablation, cost, and regime analyses, even though they are not statistically significant in the formal sense of rejecting a null hypothesis on this single window. Accordingly, the contribution of the ensemble is framed throughout the paper in terms of consistent and practically meaningful risk reduction, and we deliberately refrain from claiming statistical dominance over buy-and-hold.

M. WALK-FORWARD ROBUSTNESS

Table 14 reports the results of the anchored, three-fold walk-forward. For each fold we report the selected configuration on the training window and the test-window total return. The selected configuration is stable across folds, which indicates that the headline configuration is not an artifact of a single chronological split.

N. SENSITIVITY TO THE SELECTION OBJECTIVE

The headline protocol selects $(m^\star, c^\star)$ by in-sample total return (Section V-C), which invites the question whether a risk-adjusted objective would have selected a different configuration. To answer it, the identical training-window grid is re-ranked under two risk-adjusted objectives, the Sharpe ratio and the Calmar ratio, and the winning configuration of each is evaluated untouched on the held-out window. On WIG20, all three objectives select the same configuration, (`aggressive`, $c_{\min} = 0.05$), and the FTSE 100 replication behaves identically, selecting (`aggressive`, $c_{\min} = 0.15$) under every objective. Every out-of-sample number reported in this paper is therefore invariant to replacing the total-return objective with a risk-adjusted one on both markets. The mechanism is visible in Table 2: the *passive* configurations never trade and the *balanced* configurations trade too rarely to compete on either cumulative return or ratio statistics, so the argmax falls inside the *aggressive* column under every objective. This invariance cannot be assumed in general - it may fail on grids with more degrees of freedom or on longer menus of modes - but on the present design it confirms that the choice of a simple, low-variance selection objective (Section V-C) is not load-bearing, consistent with the estimation-noise argument for avoiding ratio objectives on short training samples [45].

O. EXTERNAL VALIDATION ON THE FTSE 100

To test whether the risk-aware profile of the ensemble is specific to the WSE or transfers to a different market structure, we replicate the entire pipeline on the FTSE 100 panel described in Section V-A. The transaction-cost model is adjusted to reflect UK conditions: a proportional commission of 5 bps and one-way slippage of 5 bps on every trade, plus a 50 bps Stamp Duty Reserve Tax charged on purchases only, which makes the FTSE cost environment asymmetric and, on the buy side, considerably more punitive than the WIG20 calibration. The decision-agent grid, the indicator hyperparameters, and every evaluation procedure are held identical; only the market data and the cost model change.

The anchored walk-forward selects the same decision-agent family as on WIG20 (`aggressive` mode) and evaluates it on 444 held-out trading days. Table 15 reports the headline comparison. As on WIG20, the ensemble trails buy-and-hold on gross total return (0.14 against 0.25) but leads on every risk axis: the Sharpe ratio is 1.37 against 1.13, the maximum drawdown is -0.07 against -0.13 (a reduction of roughly one half), and the Calmar ratio is 1.18 against 1.00. The extended risk-adjusted metrics show the same pattern: the Sortino ratio is 1.99 against 1.52, the Ulcer Index is 1.59 against 2.66, and the downside deviation is 0.04 against 0.09. The one interesting reversal is the Martin ratio (4.91 against 5.05), which is marginally lower for the ensemble because the FTSE test window is overwhelmingly a bull market and buy-and-hold sustains a very shallow integrated drawdown over it.

The regime decomposition explains the aggregate numbers. The FTSE test window is 90.32% bull (401 of 444 days), with only 5 correction days and 38 sideways days, so it offers little opportunity for the correction-regime defense that drives the WIG20 result. Even so, during the sideways subperiods the ensemble earns 0.02 against -0.03 for buy-and-hold, the same sideways-carry pattern observed on WIG20; the aggregate total-return shortfall is again concentrated in the bull regime (0.14 against 0.30), where a partially invested strategy mechanically trails a fully invested benchmark. Matching the two series on ex-ante volatility, as in Section VI-D, raises the ensemble total return to 0.26 at a leverage of $\lambda = 1.82$, a gain of $+1.87$ percentage points over buy-and-hold with a still-smaller drawdown (-0.12).

The paired bootstrap tests tell the same statistical-power story as on WIG20. The buy-and-hold point estimates lie inside the ensemble confidence intervals, and the one-sided p-values for the ensemble exceeding buy-and-hold are 0.789 on total return and 0.331 on Sharpe ratio; none crosses $p < 0.05$. The LSTM baseline posts higher point estimates than the ensemble on this particular window, both on gross total return (0.27 against 0.14) and on Sharpe ratio (1.49 against 1.37), but neither gap is statistically significant (the one-sided p-value for the ensemble exceeding the LSTM on total return is 0.879). Two observations put this apparent edge in perspective. First, it is a bull-market-exposure effect rather than evidence of superior risk control: the LSTM remains almost continuously invested across a window that is 90.32% bull, so it captures more of the upside at the price of a deeper drawdown (-0.08 against -0.07). Second, the LSTM is itself statistically indistinguishable from passive buy-and-hold on this window (one-sided $p = 0.411$), which is consistent with a near-fully-invested strategy tracking the index; the transparent ensemble, by contrast, deliberately trades part of that exposure for the shallower drawdown documented above. The GBM baseline pushes the same pattern to its extreme: it posts the highest point estimates on the window (0.35 total return at a Sharpe ratio of 1.63), and it does so by never closing a position over the 444 held-out days - an effectively fully invested stance in a window that is 90.32% bull. On total return its gap over the ensemble is marginally significant at the five-percent level (the one-sided p-value for the ensemble exceeding the GBM is 0.951), although the corresponding Sharpe-ratio comparison is not (one-sided $p = 0.682$), and its advantage over buy-and-hold remains statistically non-significant (one-sided $p = 0.144$ on total return) - the signature of a near-fully-invested strategy tracking a bull market rather than of superior risk control. Its maximum drawdown of -0.10 is deeper than the ensemble's -0.07, as the LSTM's is, so the qualitative trade-off is unchanged: the learning-based baselines capture more bull-market upside by staying exposed, while the ensemble deliberately trades upside for drawdown control. The auxiliary analyses replicate on this market as well: the weight-sensitivity experiment yields the same qualitative ranking (Section VI-J), the walk-forward selection is invariant to the objective (Section VI-N), and the confidence heuristic is equally uninformative at the signal level (Section VI-H). Taken together, the FTSE results show that the central finding of the paper - a consistent drawdown and downside-risk advantage over passive buy-and-hold, purchased with some bull-market upside - transfers to a second market with a markedly different cost structure, and is not an artifact of the WIG20 sample.

TABLE 15. External validation on the FTSE 100 held-out test window, net of transaction costs (including the 50 bps buy-side Stamp Duty Reserve Tax). Brackets give 95% bootstrap confidence intervals.

Metric	Multi-agent system	LSTM baseline	GBM baseline	Buy-and-hold FTSE
Total return	0.14 [−0.01, 0.32]	0.27	0.35	0.25
CAGR	0.08	–	–	0.13
Sharpe ratio	1.37 [−0.11, 2.94]	1.49	1.63	1.13
Max drawdown	−0.07	−0.08	−0.10	−0.13
Calmar ratio	1.18	–	1.88	1.00
Sortino ratio	1.99	–	–	1.52
Ulcer Index	1.59	–	–	2.66
Number of trades	55	37	0	–

VII. DISCUSSION

A. FROM FORECAST-COMBINATION THEORY TO EMPIRICAL EVIDENCE

Equation (1) predicts that the combined forecast error is strictly smaller than the smallest individual forecast error whenever the individual errors are not perfectly correlated. Table 8 and Fig. 5 show that the empirical pairwise correlations of the agent signals span the full range from -0.71 to 0.46, with a mean absolute off-diagonal correlation of 0.25. The theoretically most informative pairs are those with negative correlations, in particular RSI vs. ROC (-0.71) and Bollinger vs. ROC (-0.55). The RSI-ROC pair is especially diagnostic: RSI is a mean-reversion rule that emits a buy signal in oversold conditions, whereas ROC is a momentum rule that emits a sell signal in the same conditions. The two agents therefore naturally disagree, and the score of the decision agent resolves the disagreement by weighing confidences across families. In a window dominated by the bull regime, the trend family (MA trend) dominates the score; in sideways or correction regimes, the mean-reversion and volatility families gain relative weight. The regime decomposition in Table 6 shows the footprint of this mechanism: the ensemble loses less than buy-and-hold during corrections and earns more during sideways regimes, which is precisely the pattern predicted by a variance-minimizing combination of agents with heterogeneous error structures.

A concrete illustration clarifies the mechanism. Consider a day on which a WIG20 constituent has rallied sharply over the prior ten sessions. The ROC agent, a pure momentum rule, reports a buy signal with high confidence because the recent rate of change is large and positive. The RSI agent simultaneously reports a sell signal with moderate confidence because the 14-day relative strength has crossed above its 70 overbought threshold. The MA trend agent reports a buy signal, because the fast SMA is above the slow SMA, but with modest confidence because the spread between the two averages is small relative to the latest close. Under the family weights of Section IV, the combined score in (2) receives positive contributions of roughly $1.0 \cdot c_{\mathrm{ROC}}$ from momentum and $1.3 \cdot c_{\mathrm{MA}}$ from trend, and a negative contribution of $-0.8 \cdot c_{\mathrm{RSI}}$ from mean reversion. Whether the final action is a buy, a hold, or a sell depends on the confidences: on days when the RSI overbought signal is strong, the aggregated score can fall below the buy threshold even though two of the three agents call for a long position. This is precisely the variance-reduction mechanism of (1) expressed in decision-agent terms: the ensemble uses the disagreement between a momentum agent and a mean-reversion agent to step out of a position that a single-family rule would have kept open, which in turn translates into the shallower drawdowns visible in Fig. 3.

The confidence-validation experiment (Section VI-H) sharpens what this theoretical connection does and does not claim. Because the confidence values turn out to carry no measurable information about signal-level accuracy, the inverse-variance reading of (2) must be understood as the conceptual motivation for the architecture rather than as a calibrated statistical mechanism; what the data support is the weaker and more robust half of the theory, namely that pooling forecasters with imperfectly correlated errors reduces the variance of the combined position, regardless of how the pool is weighted. The weight-sensitivity results reinforce this reading from the other side: near-equal (inverse-variance) weights, equal weights, and the fixed prior all deliver similar out-of-sample performance, exactly as the forecast-combination literature predicts for combinations whose members have similar individual accuracy [20], [43]. In short, the empirical content of the theory in this application is error diversity, not precision weighting, and the paper's claims are framed accordingly.

B. EVIDENCE FOR THE THREE HYPOTHESES

The headline results, read together with the risk-adjusted table, the volatility-matched table, and the regime decomposition, are consistent with H1. On the aggregate test window, the ensemble trails buy-and-hold on gross total return by -36.84 percentage points in the bull regime, more than enough to explain the aggregate shortfall, but on point estimates it leads buy-and-hold on every risk-adjusted metric considered: Sharpe (1.06 vs. 0.82), Sortino (1.60 vs. 1.19), Calmar (2.10 vs. 0.96), Ulcer Index (2.39 vs. 6.69), Martin ratio (5.29 vs. 2.49), and maximum drawdown (-0.06 vs. -0.17). Once the two series are matched on ex-ante volatility, the ensemble also leads buy-and-hold on total return, by $+18.55$ percentage points, with materially smaller drawdown. As the significance analysis in Section VI-L makes clear, these advantages are point-estimate-level and do not attain conventional statistical significance on a single sub-two-year window, so we read them as consistent and economically meaningful rather than as statistical dominance. The FTSE 100 replication in Section VI-O yields the same qualitative H1 pattern on 444 held-out trading days under UK-specific costs, supporting external validity even though the formal hypothesis is stated against WIG20. These observations ~~instantiate~~ are consistent with the variance-reduction prediction of forecast-combination theory: the combination of heterogeneous directional forecasts lowers the variance of the equity curve, and the reduction is visible both in the volatility and in the drawdown profile.

The comparison against the best single-agent baseline supports H2 on point estimates: the ensemble total return of 0.23 exceeds the best single-agent total return of 0.16 (produced by `only_rsi`) by more than six percentage points on the test window. The bootstrap p-value of 0.268 indicates that, with a single test window, this point-estimate advantage does not pass the $p < 0.05$ threshold. However, this is consistent with the known difficulty of establishing statistical significance for Sharpe-like metrics over windows of fewer than 400 trading days [12]. Interpreted through the lens of forecast-combination theory, the positive point-estimate advantage is a direct illustration of the variance-reduction argument: the diversified score has a smaller error variance than any single forecaster and achieves a materially higher total return without a proportional increase in drawdown.

The ablation strongly supports H3. Removing any single signal agent moves the total return on the test window within a narrow band around the full ensemble, and the worst leave-one-out configuration still achieves a total return above 0.19, a Sharpe ratio above 1, and a Sortino ratio above buy-and-hold. Two observations deserve particular mention. First, removing the momentum family (MACD) slightly improves the headline metrics on this test window, which indicates that the pre-set family weights overweight momentum relative to what this particular period demands; ~~in a more conservative future version, the family weights could be learned rather than fixed~~the weight-sensitivity analysis of Section VI-J confirms this hindsight observation directly (downweighting momentum improves the window), while also showing that in-sample weight optimization does not reliably capture such regime-specific adjustments out of sample. Second, the Bollinger agent on its own produced very few trades under the selected configuration, so its removal has only a mild effect on the behavior of the ensemble; it behaves as a low-activity member of the forecast pool on this test window. Taken together, these findings show that the ensemble is robust to the removal of any single signal agent, consistent with the forecast-combination prediction that variance reduction persists as long as at least two imperfectly correlated forecasters remain. The weight-sensitivity analysis extends this robustness from the roster of agents to the weights themselves: no perturbation, reordering, or data-driven re-estimation of the family weights changes the qualitative conclusion.

C. CROSS-MARKET GENERALIZATION

The FTSE 100 replication in Section VI-O addresses the most immediate external-validity concern, namely that the risk-aware profile of the ensemble might be an artifact of the WIG20 sample. The qualitative conclusion is unchanged on a market with a different constituent set, a different currency, and a materially different cost structure that includes an asymmetric 50 bps buy-side transaction tax: the ensemble trails passive buy-and-hold on gross total return during a bull-dominated window, yet delivers a higher Sharpe ratio (1.37 against 1.13), a shallower maximum drawdown (-0.07 against -0.13), and a better Sortino ratio and Ulcer Index, and it overtakes buy-and-hold on total return once the two series are matched on ex-ante volatility. That the same trade-off appears on two markets, despite an added stamp-duty friction that penalizes the more active strategy, supports the interpretation that the drawdown and downside-risk advantage is a structural consequence of forecast-combination rather than a market-specific coincidence. The transfer is not perfect: the FTSE test window contains almost no correction days, so the correction-regime defense that is prominent on WIG20 has little room to act, and the integrated-drawdown advantage narrows to the point where the Martin ratio marginally favors buy-and-hold. This is consistent with the regime-dependence predicted by the adaptive markets hypothesis [42]: the ensemble adds the most value in the regimes it is designed to defend, and a window that is almost entirely bull is the least favorable setting for it.

Beyond the two markets studied here, three considerations delimit how far the results should be extrapolated. First, the profitability of technical trading rules varies systematically across markets and erodes as markets mature: survey evidence reports stronger rule profitability in emerging and less liquid markets and weakening evidence in developed ones once transaction costs and data-snooping adjustments are applied [49], and classic evidence from Asian markets ties rule profitability to the level of market development and trading frictions [50]. On this spectrum, the present results are consistent with two moderately mature markets in which naive signal-chasing no longer pays - every single-agent baseline is unprofitable or marginal, and neither learned baseline achieves a statistically significant edge - but a disciplined, diversified risk overlay still adds risk-adjusted value. We would accordingly expect the gross-return picture to be more favorable to the ensemble in less efficient, higher-friction markets, and the risk-control profile to be the more portable component in highly developed ones. Second, the transferable part of the framework is structural rather than numerical. The agent interfaces, the anchored walk-forward protocol, the calibrated cost model, and the evaluation battery apply unchanged to any market with reliable daily closes - the FTSE replication required only a new data source and cost model - whereas the specific calibrations (thresholds, weights, commissions, taxes) must be re-derived per market, as the asymmetric UK stamp duty illustrates. Third, the value added by the ensemble is regime-dependent, concentrated in corrections and sideways phases, so its realized advantage in any market will track the regime composition of the evaluation window; markets or periods dominated by persistent bull regimes mechanically favor passive exposure, as both test windows here illustrate. Mapping the framework's domain of validity across further developed and emerging indices, as outlined in Section VIII, is therefore the most direct extension of this study.

D. WHY GROSS TOTAL RETURN UNDERSTATES THE CONTRIBUTION OF THE ENSEMBLE

The standard comparison of a trading strategy against buy-and-hold reports a single gross total return for each series and declares the larger number the winner. On the present test window that comparison would declare the passive benchmark the winner, even though the equity curve of the ensemble has less than half of the drawdown of the benchmark and delivers materially higher Sharpe, Sortino, Calmar, Ulcer, and Martin ratios. Three observations suggest that the gross-total-return comparison is the wrong summary in this setting. First, the ensemble is not fully invested on all dates by design: the decision-agent thresholds ensure that positions are opened only when the combined score crosses a confidence level. In a test window that is 52.87% bull, any partially-invested strategy will trail a fully-invested benchmark on gross cumulative return almost by construction. Second, the volatility-matched comparison in Section VI-D shows that at equal ex-ante risk budget the ensemble produces $+18.55$ percentage points more total return than buy-and-hold, with a smaller drawdown in absolute terms. Third, the regime decomposition in Section VI-E shows that the aggregate shortfall of the ensemble is entirely attributable to the bull regime; in the correction and sideways regimes, the ensemble matches or beats buy-and-hold even on gross return. These three pieces of evidence reframe the comparison: the ensemble adds value as a *risk-aware decision support tool* that trades some bull-regime upside for shallower drawdowns and higher risk-adjusted performance overall.

E. CONNECTIONS TO THE BROADER EMPIRICAL LITERATURE

The pattern of results reported in Section VI is consistent with three well-documented stylized facts of empirical finance. First, time-series momentum and cross-sectional momentum are empirically orthogonal, and portfolios that combine them deliver better risk-adjusted performance than either alone [23]–[25]. The negative correlation, within the ensemble, between the momentum family (MACD, ROC) and the mean-reversion family (RSI) is the signal-level analog of this finding: the two families produce forecasts that tend to disagree precisely when one or the other would have been wrong on its own, and the score of the decision agent integrates the two into a more conservative directional bet than either family would produce alone. Second, the literature on the adaptive markets hypothesis of [42] predicts that no single trading rule can dominate across all regimes, because the efficiency of a market with respect to any given rule is itself time-varying. The regime decomposition in Section VI-E is consistent with this prediction: the trend-heavy ensemble loses ground during the bull regime, where a passive benchmark mechanically dominates any partially-invested strategy, but it outperforms the passive benchmark during corrections and sideways periods, where mean-reversion and volatility agents contribute directional information that the trend agent on its own would miss. Third, the literature on risk-adjusted performance evaluation has long argued that the Sharpe ratio alone is inadequate for comparing strategies with different drawdown profiles [39]–[41]. The quantitative spread between the Sharpe advantage of the ensemble (1.06 vs. 0.82, a factor of 1.3) and its Martin-ratio advantage (5.29 vs. 2.49, a factor of 2.1) illustrates the difference between adjusting for realized volatility and adjusting for the full drawdown profile, and explains why the paper leads with Calmar and Ulcer in its headline claim rather than with Sharpe alone.

F. PRACTICAL IMPLICATIONS FOR A RETAIL INVESTOR

For a WSE-focused retail investor with access to standard daily data and typical Polish brokerage conditions, two practical implications stand out. First, the ensemble reduces the worst realized drawdown by a factor greater than two and reduces the integrated drawdown, as measured by the Ulcer Index, by nearly two-thirds. This is the dominant axis on which retail portfolios tend to be evaluated in practice, both because maximum drawdown has a direct operational meaning (it is the worst unrealized loss the investor has to live through) and because sustained drawdowns are empirically associated with behavioral deviations from the original plan. Second, the cost-sensitivity sweep shows that the risk-adjusted edge of the ensemble over buy-and-hold is preserved at every plausible combination of commission and slippage, including the punitive 100 bps / 20 bps configuration, which is well above current WSE retail tariffs. Neither of these implications depends on leverage, and both can be consumed by a retail investor as decision-support signals overlaid on a passive benchmark rather than as a replacement for the benchmark.

Two operational considerations follow from this framing. The first concerns the update cadence. The five agents and the decision agent consume only end-of-day closes, so a retail deployment amounts to running the pipeline once after the daily WSE close and acting on the resulting recommendations at the open or close of the following session. No intraday infrastructure, tick data, or order-book integration is required, which lowers the bar for adoption outside the institutional segment. The second concerns monitoring. Because the decision-agent score in (2) is a linear combination of agent signals weighted by confidence and family, the score can be decomposed, day by day, into per-agent contributions and plotted alongside the equity curve. This decomposition gives the investor an interpretable audit trail of which agents drove each position change, and makes it straightforward to detect structural shifts in the signal distribution (for example, an agent whose confidence collapses for an extended period) without retraining the ensemble. We view this interpretability as a distinctive strength of a ~~theory-grounded~~ theory-motivated forecast-combination design relative to end-to-end deep-reinforcement-learning trading systems, which typically produce a scalar action without a transparent attribution to individual input signals. This strength ~~does not come at the cost of~~ comes at little cost in performance on the markets studied here~~: the LSTM directional baseline~~. On WIG20 the ensemble leads both the LSTM and GBM directional baselines of Section V-D1 ~~, a representative~~ - two structurally different learning-based ~~competitor~~ competitors trained on the same price history and executed under identical costs ~~, does not beat the transparent ensemble by a statistically significant margin on either market. On WIG20 the ensemble dominates it on every metric~~ - on total return and on every risk-adjusted ratio, notwithstanding the marginally shallower drawdown that the GBM achieves by barely taking any risk at all (Section VI-K)~~; on~~. On the FTSE 100 the ~~LSTM posts~~ learned models post higher point estimates, ~~but~~ and the GBM's total-return gap is marginally significant on that single window, but they achieve this only by remaining near-fully invested through an almost pure bull window, which leaves ~~it~~ them statistically indistinguishable from passive buy-and-hold while giving up the drawdown control that motivates the ensemble (Section VI-O). The ensemble thus retains full per-signal attribution while remaining competitive ~~with an opaque learned model~~on the risk axis with opaque learned models.

G. DEPLOYMENT AND SCALABILITY

Several practical considerations determine whether the system can be deployed beyond a backtest, and the transparent design has favorable properties on most of them. *Computational cost.* Decision-time inference is non-iterative: each signal agent is a closed-form indicator computed over a rolling window, updated incrementally per symbol, and the decision agent reduces the per-symbol signals to a single confidence-weighted sum followed by a threshold comparison, as in (2). There is no model training, no gradient computation, and no specialized hardware at decision time; one pass over the day's closing prices on a commodity CPU suffices. This contrasts with the LSTM ~~baseline~~ and GBM baselines of Section V-D1 and with end-to-end deep-reinforcement-learning systems, which require training and periodic retraining. *Scalability across symbols and markets.* The total work per day is of order $O(N_{\text{symbols}} \times N_{\text{agents}})$, linear in the size of the traded universe, and it is embarrassingly parallel because each agent processes the history of a single symbol independently through the shared base-agent interface. The FTSE 100 study of Section VI-O is concrete evidence of market portability: re-pointing the entire pipeline at a different exchange is a configuration change, comprising a new data source and cost model, rather than a change to the agents or the decision layer. *Liquidity.* The evaluation trades only large-capitalization index constituents at the daily close, and the selected configuration is low-turnover, so the resting-order footprint is small at retail size; price impact, partial fills, and order-book dynamics are not modeled and are discussed as a limitation in Section VIII. *Execution latency.* Because signals are computed once after the close and acted on at the next session, the design tolerates latencies of seconds to minutes and requires no co-location, tick data, or order-book integration, which places it at the opposite end of the infrastructure spectrum from high-frequency trading. *Portfolio scalability and the retail-institutional distinction.* The fractional position-sizing rule scales capital mechanically, but the assumption of fills at the closing price without impact bounds the deployable capital relative to average daily volume, so the profile is most defensible at retail-to-moderate assets under management. For a retail investor the system is usable as an end-of-day decision-support overlay on a passive benchmark, without leverage or dedicated infrastructure; for an institutional user the linear scaling and market-agnostic interface make wider universes feasible, but explicit price-impact modeling and the analytical-only leverage caveat of Section VI-D would become binding.

H. DESIGN IMPLICATIONS

Three design implications follow from the experiments. First, the transparency of the decision-agent score, which is a linear combination of agent signals weighted by confidence and by family, is both a theoretical convenience and a practical one. The contribution of each agent on a given day can be read off directly from the score, and the ablation shows that this attribution is consistent with the marginal contribution of that agent to the headline metrics. Second, the fact that the cost-sensitivity sweep in Table 7 preserves a Sharpe ratio strictly above the passive benchmark at every plausible cost configuration suggests that the ensemble is not surviving only thanks to a favorable cost calibration. Third, the stability of the selected (mode, $c_{\min}$) configuration across the three walk-forward folds indicates that a simple grid-search procedure on a long enough training window is adequate, and that more elaborate parameter-adaptation machinery is not needed at the daily frequency on WIG20.

VIII. LIMITATIONS AND FUTURE WORK

Several limitations of this study should be kept in mind. First, although the evaluation spans two markets (the WIG20 and, as an external validation, the FTSE 100), both are large-capitalization blue-chip indices evaluated at daily frequency; generalization to smaller and less liquid stocks, to other asset classes, or to intraday frequencies would require a separate evaluation, and the signal agents and the decision agent, which were designed around daily closes, would need to be adapted for intraday use. Second, although the transaction-cost model captures commissions and slippage and is stress-tested over a wide grid, it does not capture liquidity constraints, partial fills, or the price impact of larger orders. In practice, large orders could move the closing price and would require an order-book-level simulation. Third, the system does not use fundamental data, news, or alternative data sources, which are known to carry orthogonal information with respect to price-based indicators and would fit naturally into the forecast-combination framework as additional agents with distinct error structures. Fourth, the decision agent uses fixed family weights ~~, which could in principle be learned from data; we chose fixed weightsto keep the degrees of freedom low and the interpretation of the score transparent~~and heuristic confidence values. The sensitivity analysis of Section VI-J shows that the headline conclusions are robust to perturbing, reordering, or re-estimating the weights, and that in-sample optimization does not reliably improve on the fixed prior; nevertheless, the weights are static over time, and regime-conditional weighting remains untested. Relatedly, Section VI-H shows that the confidence values do not predict signal-level accuracy, so the decision layer cannot claim calibrated precision weighting and the link to classical forecast-combination theory remains conceptual. Fifth, the test window, although long enough to cover bull, correction, and sideways regimes, is ultimately a single window of under two years, and the bootstrap tests have limited power to detect moderate effect sizes on a sample of this size; we therefore rely on point estimates and confidence intervals as much as on p-values. Sixth, the volatility-matched comparison requires leverage of $\lambda = 2.06$, which is impractical for a typical retail investor on the WSE; the comparison therefore serves as a counterfactual on risk budget rather than as an operational recommendation. Finally, the paper does not include a live-trading exercise; any claims about realized performance in production would require such an exercise.

Each of these limitations maps to a specific direction for future work. Building on the FTSE 100 validation reported here, the system could be extended to further indices (for example, the German DAX, the Italian FTSE MIB, or the US S&P 500) and to smaller-capitalization or emerging-market segments to map more precisely the market conditions under which the risk-adjusted advantage holds. Intraday frequencies would require a careful revision of both the signal agents and the transaction-cost model, and would expose the system to additional market-microstructure effects [51]. Fundamental and alternative data could be integrated as additional signal agents that share the same base-agent interface, which is straightforward given the modular design. ~~Learned family weights, for example via a Bayesian update of the~~ Because static in-sample weight optimization and inverse-variance ~~proxy, would formalize the link to the forecast-combination argument of Section III. Regime-conditional~~ estimation add little (Section VI-J), the more promising weighting extension is regime-conditional family weights, in which the trend family is upweighted in bull regimes and downweighted in corrections~~, are a natural extension~~; this is a natural evolution of the current fixed-weight design and would be expected to recover part of the bull-regime shortfall identified in Section VI-E. A complementary direction is to replace the heuristic confidence values with calibrated probabilities, for example by isotonic or Platt-type mappings from indicator magnitudes to empirical hit rates estimated on the training window [48]; calibrated confidences would let the decision layer implement genuine precision weighting and would upgrade the conceptual link to forecast-combination theory (Section VI-H) into a testable quantitative one. Beyond weighting, the fixed roster of agents could be made adaptive through dynamic agent selection, in which the active pool is chosen per regime or per symbol according to recent forecast accuracy, so that agents contribute only where they have demonstrated skill. More ambitiously, the linear confidence-weighted score could be replaced by a learning-based ensembling layer, such as a stacked meta-learner or a gating network that learns the combination function from data; the challenge there is to retain the per-agent attribution and interpretability that motivate the present design while admitting a richer, potentially non-linear combiner. Finally, a walk-forward live-trading exercise on a paper-trading broker interface would close the loop between the backtest and realistic execution.

IX. CONCLUSION

The primary contribution of this paper is a rigorous and reusable evaluation framework, together with an in-depth empirical study of a transparent, theoretically ~~grounded~~ motivated ensemble that instantiates it. Read this way, the paper makes methodological, empirical, and theoretical claims about risk-aware multi-agent decision support on less-studied equity markets.

Methodologically, the evaluation pipeline - anchored walk-forward tuning of decision-agent parameters, calibrated frictions on every trade, ablation across single-agent removals, ~~a comparison against an LSTM deep-learning baseline~~sensitivity analyses over the family weights and the selection objective, an empirical audit of the confidence heuristic, comparisons against LSTM and gradient-boosted-tree baselines, regime decomposition, a wide cost-sensitivity sweep, and bootstrap significance tests - closes the four gaps identified in Section I and is reusable as a template for similar studies on smaller equity markets, where capital preservation in turbulent periods may matter more than chasing the next bull leg.

Empirically, the ensemble does not maximize total return on the WIG20 test window; buy-and-hold does, by virtue of a bull-dominated subperiod. Once the two series are matched on ex-ante risk, however, the ensemble earns $+18.55$ percentage points more total return than buy-and-hold and improves every risk-adjusted metric reported, although these differences do not reach statistical significance on a single window. The regime decomposition shows where this comes from: $+12.57$ percentage points of correction-regime defense and $+5.14$ percentage points of sideways-regime carry, paid for by an underperformance in the bull phase that the fixed-weight design cannot fully recover. The ablation and cost-sensitivity sweep both indicate that this profile is not an artifact of a single agent or of a particular friction calibration, and the external validation on the FTSE 100 reproduces the same qualitative pattern under a different market structure and cost regime, indicating that the finding is not specific to the WSE sample.

Theoretically, the system supplies ~~an~~ a deliberately qualified empirical instance of the forecast-combination argument on an actual market~~: the~~. The pairwise correlations of the five agents span -0.71 to 0.46, satisfying the non-identical-errors assumption ~~that motivates~~ behind the variance-reduction prediction in (1)~~. The~~, and the observed risk reduction is ~~therefore not opportunistic but consistent with the theoretical baseline of Section III~~consistent with that prediction. At the same time, the confidence heuristic does not behave as a calibrated inverse-variance estimate (Section VI-H), so the connection to classical forecast-combination theory is conceptual: it correctly anticipates the benefit of pooling heterogeneous signals, while its precision-weighting refinement finds no support in these data - a distinction the paper states explicitly rather than leaving implicit.

The directions outlined in Section VIII - cross-market generalization, intraday frequencies, fundamental and alternative data, ~~learned and~~ regime-conditional family weights and calibrated confidence estimates, and a live paper-trading exercise - are the natural extensions to leverage these contributions in the next iteration of the system.

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